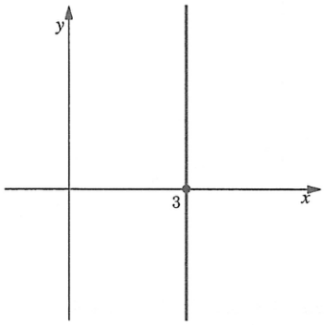
# Graphs and Transformations

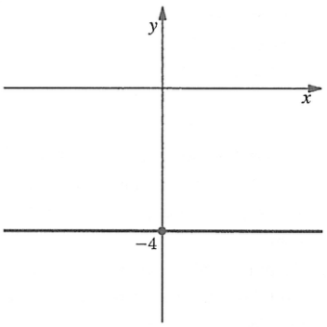
Questions 1–225

**1.04 — Question 1.** Find the equation of each of the linear graphs below:

**a)**

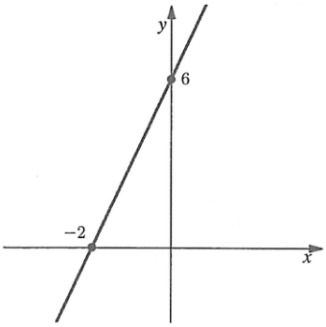



**b)**

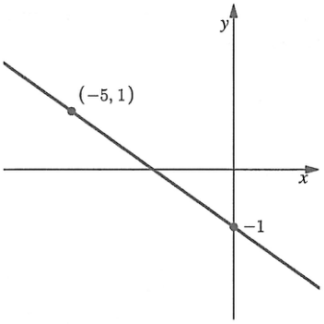



**1.04 — Question 2.** Find the equation of each of the following graphs in the form $ax + by = c$, where $a$, $b$ and $c$ are integers.

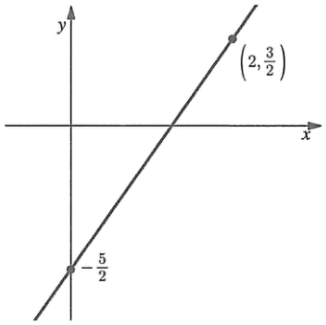
**a)**



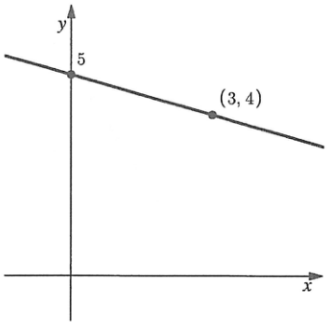

**b)**




**c)**




**d)**



**1.04 — Question 3.** Sketch the following linear graphs, identifying where they intersect the coordinate axes:

a) $y = 2x - 7$

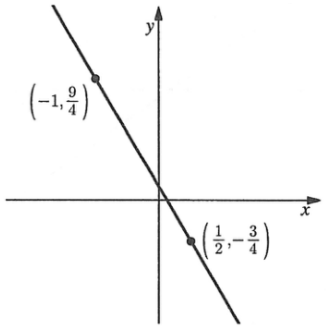
b) $y = 6 - 10x$

c) $y = \frac{1}{2}x + 5$

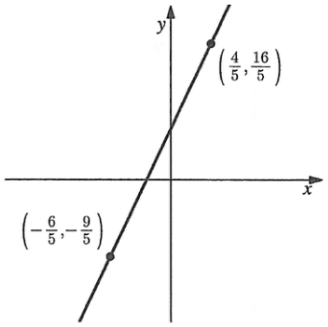
d) $y = 2 - \frac{1}{3}x$

**1.04 — Question 4.** Find the equation of each of the following graphs in the form $ax + by = c$, where $a$, $b$ and $c$ are integers.

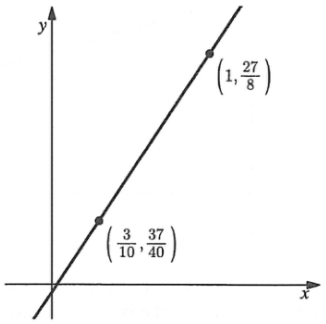
**a)**



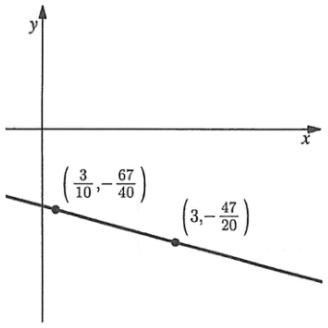

**b)**




**c)**




**d)**



**1.04 — Question 5.** Sketch the following linear graphs, identifying where they intersect the coordinate axes:

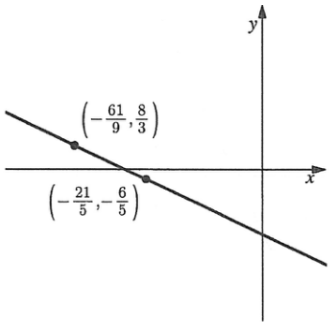
a) $2x + 5y = 10$
b) $4x - 9y = 36$
c) $-7x + 2y = 28$
d) $6x + 11y = -2$

**1.04 — Question 6.**

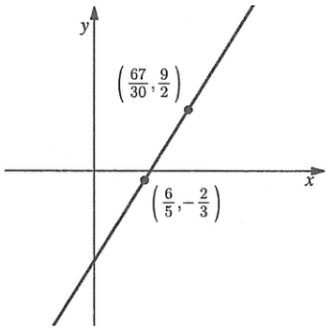
a) Sketch the graphs of $y = 2x + 8$ and $2y = 10 - x$ on the same axes, identifying where the graphs cross the coordinate axes.
b) Determine where the two lines intersect.
c) Find the area of the triangle formed by the two lines and the $x$-axis.
d) Find the area of the triangle formed by the two lines and the $y$-axis.

**1.04 — Question 7.** Find the equation of each of the following graphs in the form $x + ay = b$, where $a$ and $b$ are constants.

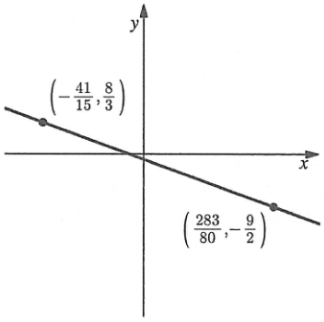
**a)**



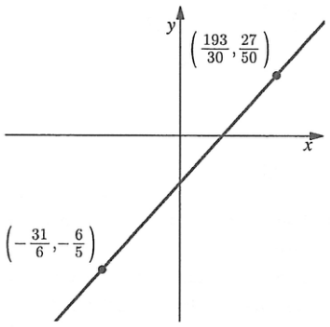

**b)**




**c)**



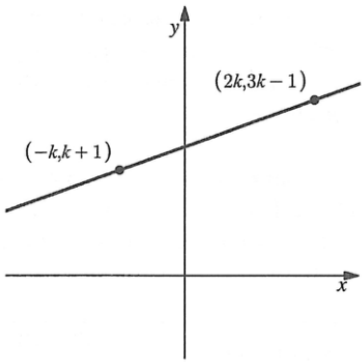

**d)**




**1.04 — Question 8.** Sketch the following linear graphs, identifying where they intersect the coordinate axes:

a) $2y = 3x - 7$
b) $5y + 1 = 6x$
c) $3y - 3 = 8x$
d) $10y = 4 - 7x$

**1.04 — Question 9.** The diagram below shows a linear function. Find the equation of the line in the form $y = ax + b$, where $a$ and $b$ are given in terms of $k$, where $k > 1$.




**1.04 — Question 10.** Sketch the following linear graphs, identifying where they intersect the coordinate axes. It is given that $k > 0$.

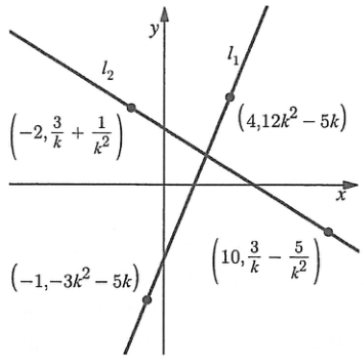
a) $y = kx + 2$
b) $y = 8 - 2kx$
c) $ky = 3 - x$
d) $2x + 7y = k$
e) $3x + ky = 9$
f) $6kx - y = -3$
g) $\frac{x}{k} + \frac{y}{2k} = 4$

**1.04 — Question 11.** The diagram below shows two lines, $l_1$ and $l_2$.

a) Find the equation of the line $l_1$.

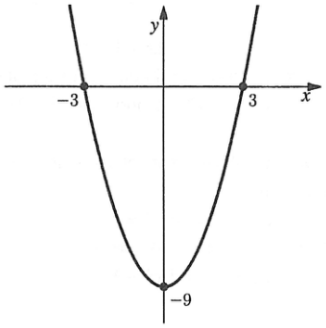
b) Find the equation of the line $l_2$.

c) Determine the coordinates, in terms of $k$, of where $l_1$ and $l_2$ intersect.

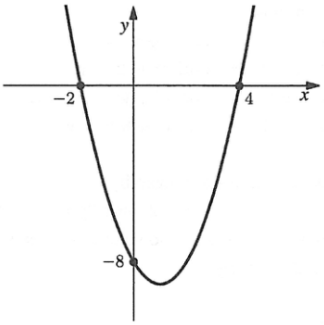



**1.08 — Question 1.** Find the equation of each of the following graphs in the form $y = (x + a)(x + b)$

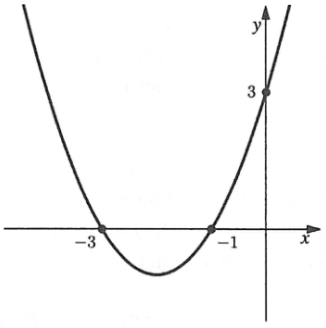
**a)**




**b)**




**c)**



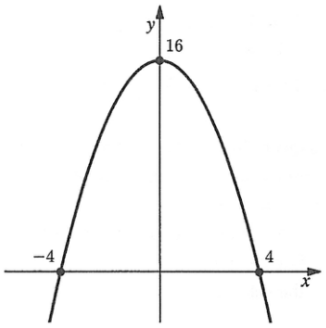
**1.08 — Question 2.** Sketch the following graphs, showing where they intersect the coordinate axes:

a) $y = (x - 5)(x - 7)$

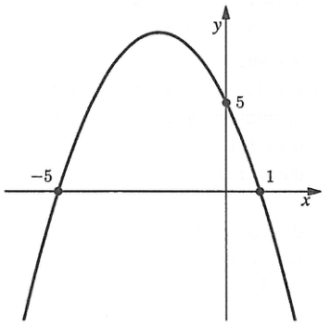
b) $y = (x + 4)(x - 2)$

**1.08 — Question 3.** Find the equation of each of the following graphs in the form $y = -(x + a)(x + b)$

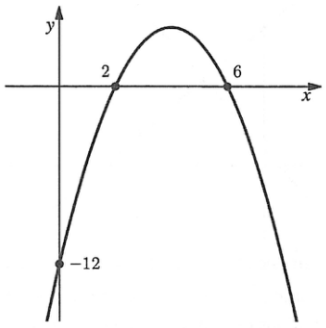
**a)**




**b)**




**c)**



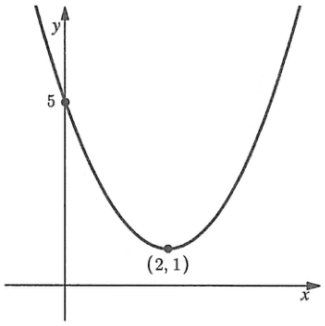
**1.08 — Question 4.** Sketch the following graphs, showing where they intersect the coordinate axes:

a) $y = -(x+2)(x-5)$

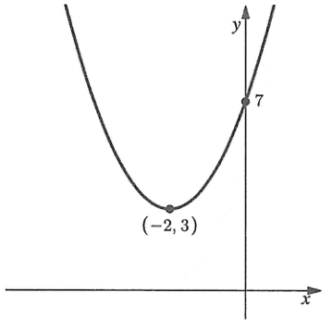
b) $y = -(x+8)(x+1)$

**1.08 — Question 5.** Find the equation of each of the following graphs in the form $y = (x+a)^2 + b$

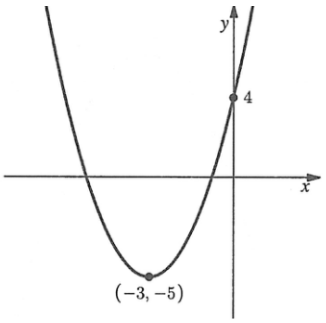
**a)**




**b)**




**c)**



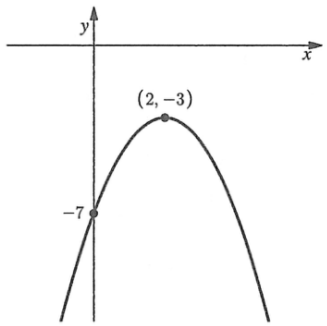
**1.08 — Question 6.** Sketch the following graphs, showing the coordinates of the turning point and where they intersect the y-axis:

a) $y = (x-4)^2 + 5$

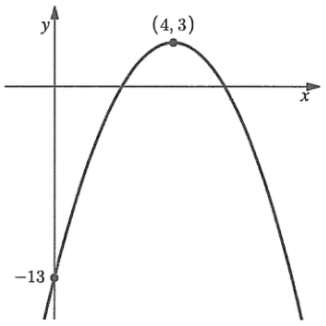
b) $y = (x-5)^2 - 6$

**1.08 — Question 7.** Find the equation of each of the following graphs in the form $y = -(x+a)^2 + b$

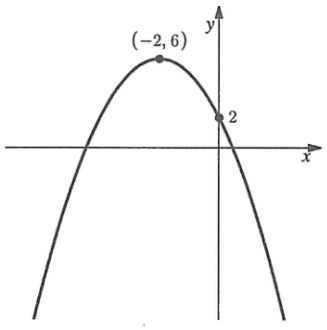
**a)**




**b)**




**c)**



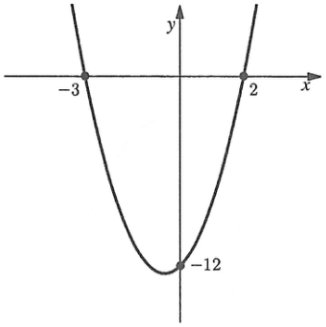
**1.08 — Question 8.** Sketch the following graphs, showing the coordinates of the turning point and where they intersect the y-axis:

a) $y = -(x+6)^2 + 2$

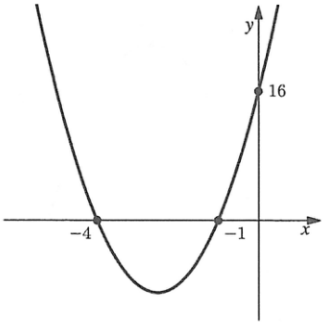
b) $y = -(x-3)^2 - 4$

**1.08 — Question 9.** Find the equation of each of the following graphs in the form $y = a(x+b)(x+c)$

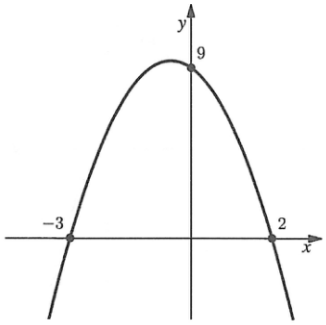
**a)**




**b)**




**c)**



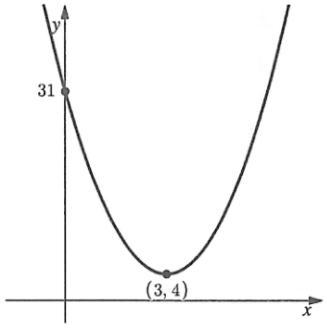
**1.08 — Question 10.** Sketch the following graphs, showing where they intersect the coordinate axes:

a) $y = 3(x-1)(x+2)$

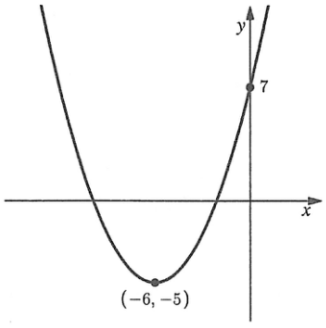
b) $y = -2(x+4)(x+3)$

**1.08 — Question 11.** Find the equation of each of the following graphs in the form $y = a(x+b)^2 + c$

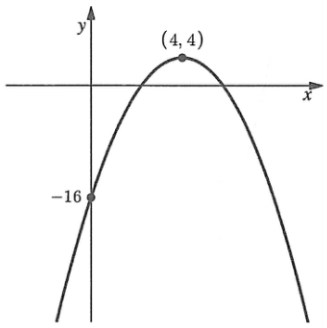
**a)**




**b)**




**c)**



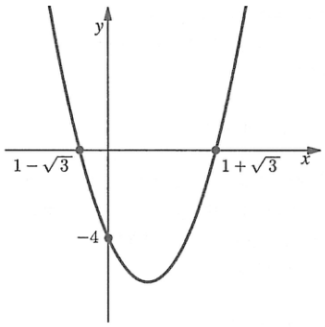
**1.08 — Question 12.** Sketch the following graphs, showing the coordinates of the turning point and where they intersect the y-axis:

a) $y = 5(x+1)^2 - 2$

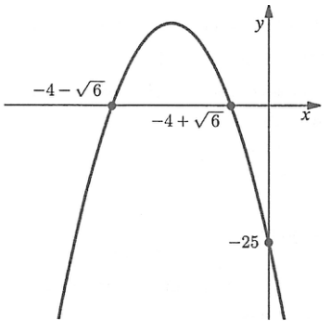
b) $y = -\frac{3}{2}(x-6)^2 - 12$

**1.08 — Question 13.** Find the equation of each of the following graphs in the form $y = ax^2 + bx + c$

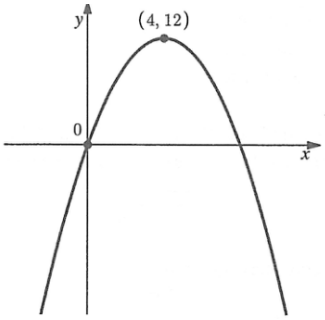
**a)**




**b)**




**c)**



**1.08 — Question 14.** Sketch the following graph, showing the coordinates of the turning point and where it intersects the $y$-axis: $y = \frac{1}{2}(x+8)^2 - 32$.

**1.08 — Question 15.** Sketch the following graphs, showing the coordinates of the turning point and where they intersect the coordinate axes:

a) $y = (x+2)(x-8)$
b) $y = (x-6)(x+4)$
c) $y = 3(x+6)(x+2)$
d) $y = -4(x-2)(x-4)$

**1.08 — Question 16.** Sketch the following graphs, showing the coordinates of the turning point and where they intersect the coordinate axes:

a) $y = (x-3)(x-7)$
b) $y = \frac{1}{3}(x+2)(x+6)$
c) $y = -(x+9)(x+5)$
d) $y = -\frac{7}{4}(x-8)(x+4)$

**1.08 — Question 17.** Sketch the following graphs, showing the coordinates of the turning point and where they intersect the coordinate axes:

a) $y = (x-6)^2 - 2$
b) $y = 8 - (x+2)^2$
c) $y = \frac{4}{3}(x+6)^2 - 9$
d) $y = -\frac{8}{3}(x-3)^2 + 12$

**1.08 — Question 18.** Sketch the following graphs, showing the coordinates of the turning point and where they intersect the coordinate axes:

a) $y = 3x^2 - 9x + 2$
b) $y = -2x^2 + 10x + 2$
c) $y = \frac{1}{2}x^2 - 6x + 9$
d) $y = -\frac{1}{5}x^2 + 2x + 10$

**1.08 — Question 19.** Given that $k > 0$, sketch the following graphs, identifying the turning point and where they cross the coordinate axes in terms of $k$:

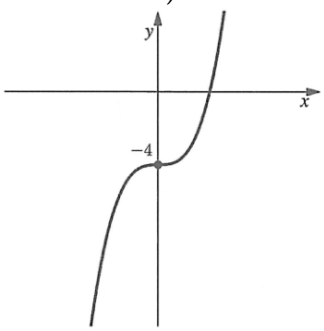
a) $y = k(2x + k)(x - 2k)$
b) $y = k - \frac{1}{3}(x - 3k)^2$
c) $y = 10 + \frac{2}{k}(kx + 1)^2$
d) $y = (3kx + 2)(3k - x)$

**1.08 — Question 20.** Given that $k > 0$, sketch the graphs of $y = (k - x)(3x + k)$ and $y = k - x$ on the same axes, showing clearly where the two graphs cross the coordinate axes and where they intersect in terms of $k$.

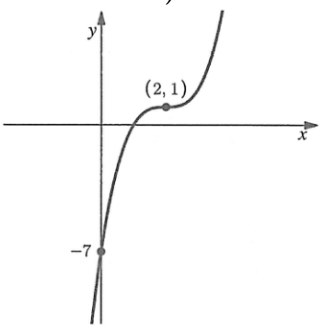
**1.15 — Question 1.** The following graphs are all functions that can be written in the form $y = (x - p)^3 + q$. Find the equation of the curve in the form

$$y = x^3 + bx^2 + cx + d$$

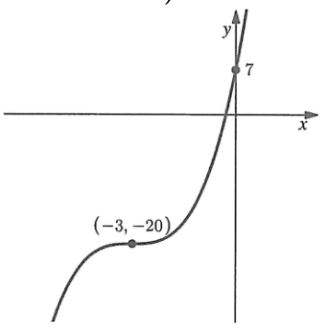
**a)**




**b)**




**c)**



**1.15 — Question 2.** Sketch the following graphs:

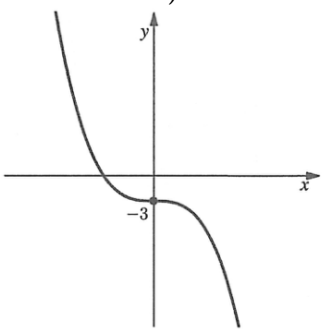
a) $y = (x+5)(x+3)(x+1)$

b) $y = (x-5)(x-3)(x-1)$

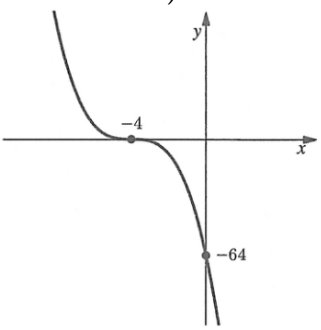
**1.15 — Question 3.** The following graphs are all functions that can be written in the form $y = q - (x-p)^3$. Find the equation of the curve in the form

$$y = -x^3 + bx^2 + cx + d$$

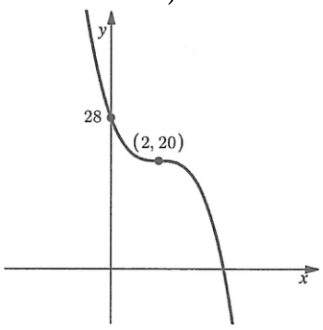
**a)**




**b)**




**c)**



**1.15 — Question 4.** Sketch the following graphs:

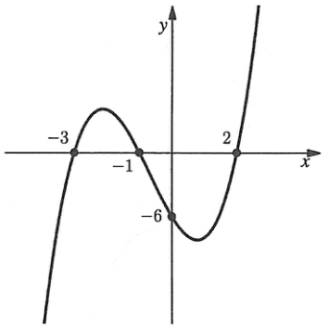
a) $y = -(x+5)(x+3)(x+1)$

b) $y = -(x-5)(x-3)(x-1)$

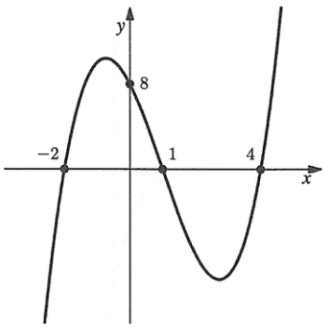
**1.15 — Question 5.** Find the equation of each of the following graphs in the form

$$y = (x-a)(x-b)(x-c)$$

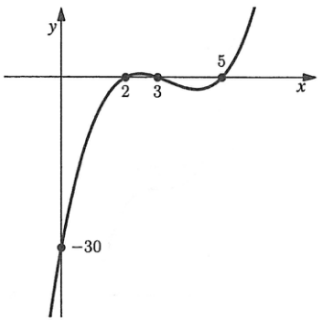
**a)**




**b)**




**c)**



**1.15 — Question 6.** Sketch the following graphs:

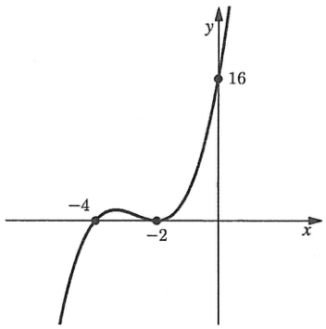
a) $y = (x+2)(x-2)^2$

b) $y = (x-2)(x+2)^2$

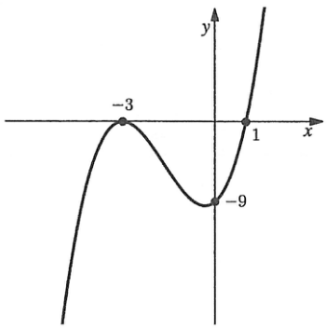
**1.15 — Question 7.** Find the equation of each of the following graphs in the form

$$y = (x-a)(x-b)^2$$

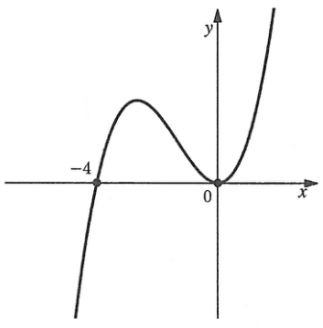
**a)**




**b)**




**c)**



**1.15 — Question 8.** Sketch the following graphs:

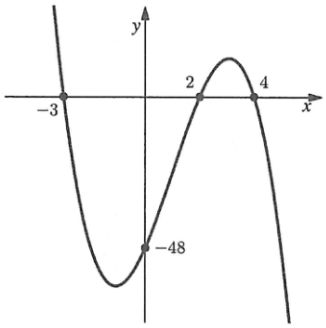
a) $y = -(x+3)(x-3)^2$

b) $y = (3-x)(x+3)^2$

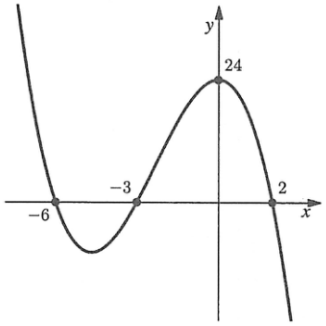
**1.15 — Question 9.** Find the equation of each of the following graphs in the form

$$y = k(x-a)(x-b)(x-c)$$

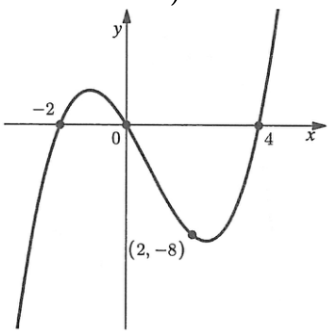
**a)**




**b)**




**c)**



**1.15 — Question 10.** Sketch the following graphs:

a) $y = \frac{1}{3}(x-6)(x-3)(x+3)$

b) $y = -\frac{3}{4}(x+8)(x+4)(x-4)$

**1.15 — Question 11.** Sketch the following graphs, showing where they intersect the coordinate axes:

a) $y = (2x+1)(x-2)(2x-7)$

b) $y = (3x-1)(2x-3)^2$

c) $y = x(4x-1)^2$

d) $y = x^2(4x-1)$

**1.15 — Question 12.** Sketch the following graphs, showing where they intersect the coordinate axes:

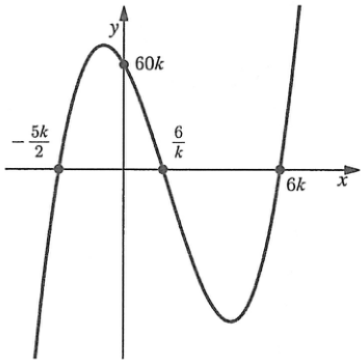
a) $y = -(x+3)(2x-1)(2x-7)$
b) $y = (4-x)(3x+2)^2$
c) $y = \frac{1}{5}(x+5)^2(5x-4)$
d) $y = -\frac{5}{3}(x+9)(x-3)(x-6)$

**1.15 — Question 13.** Given that $k > 0$, sketch the following graphs:

a) $y = (x-k)(x-2k)(x-3k)$
b) $y = (x+k)(x+2k)(x-3k)$
c) $y = (k-x)(x-2k)(x+3k)$
d) $y = (k-x)(2k-x)(x+3k)$

**1.15 — Question 14.** Given that $k$ is a positive constant, find the equation of the following graph in the form

$$y = a(bx + c)(dx + e)(fx + g)$$




**1.15 — Question 15.** Given that $k > 0$, sketch the following graph:

$$y = k(2k - x)(x - k)^2$$

**1.15 — Question 16.** Given that $k > 0$, sketch the graphs of

$$y = (x - 2k)(3kx + 20)$$

and

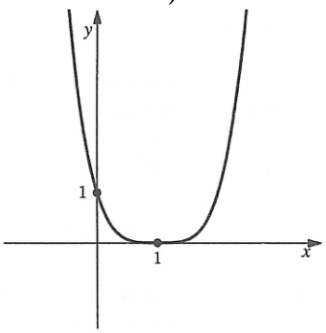
$$y = (x - 2k)(3kx + 20)x$$

on the same axes, showing clearly where the two graphs cross the coordinate axes and each other.

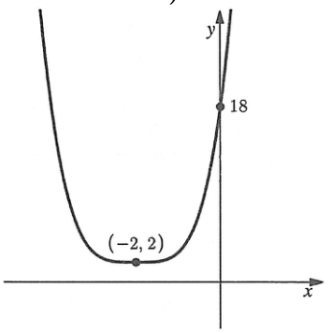
**1.16 — Question 1.** The following graphs are all functions that can be written in the form $y = (x - p)^4 + q$. Find the equation of the curve in the form

$$y = x^4 + bx^3 + cx^2 + dx + e$$

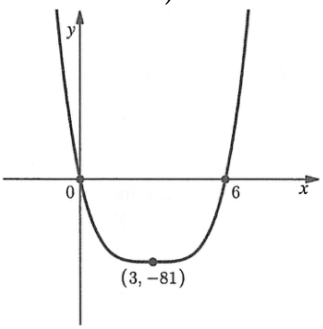
**a)**




**b)**




**c)**



**1.16 — Question 2.** Sketch the following graphs:

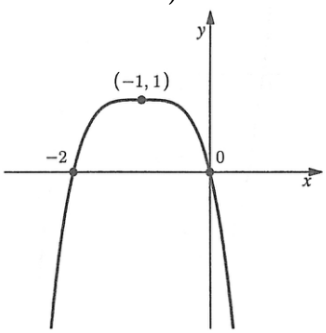
a) $y = (x-2)(x-1)(x+1)(x+2)$

b) $y = (x-3)(x+1)(x-1)(x+4)$

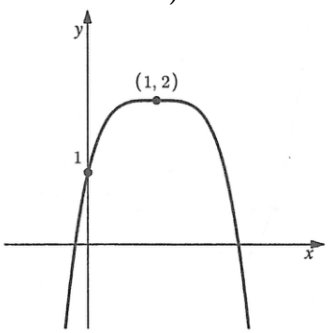
**1.16 — Question 3.** The following graphs are all functions that can be written in the form $y = q - (x-p)^4$. Find the equation of the curve in the form

$$y = -x^4 + bx^3 + cx^2 + dx + e$$

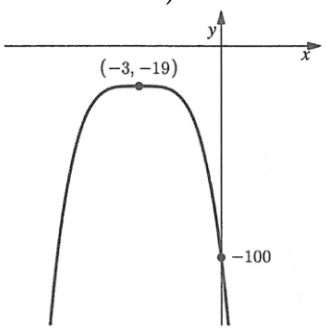
**a)**




**b)**




**c)**



**1.16 — Question 4.** Sketch the following graphs:

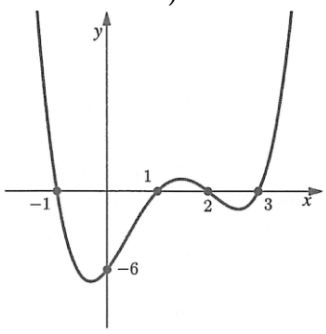
a) $y = -(x-3)(x+1)(x+2)(x+3)$

b) $y = -(x+4)(x+3)(x+2)(x+1)$

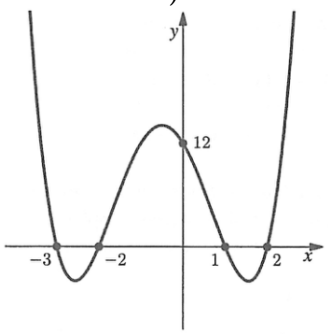
**1.16 — Question 5.** Find the equation of each of the following graphs in the form

$$y = (x-a)(x-b)(x-c)(x-d)$$

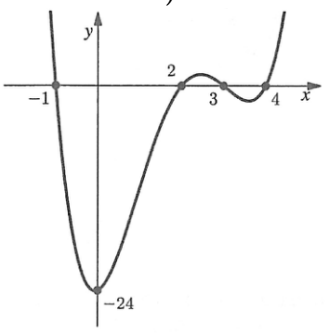
**a)**




**b)**




**c)**



**1.16 — Question 6.** Sketch the following graphs:

a) $y = (x-1)(x+3)(x-2)^2$

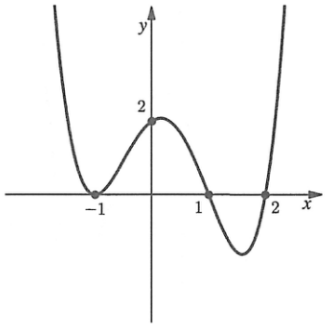
b) $y = (x+1)^2(x+2)(x-3)$

c) $y = (x+1)^2(x-2)^2$

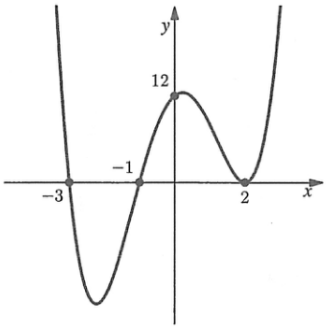
**1.16 — Question 7.** Find the equation of each of the following graphs in the form

$$y = (x-a)(x-b)(x-c)^2$$

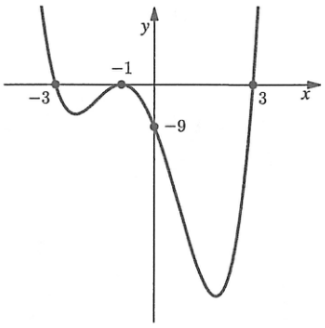
**a)**




**b)**




**c)**



**1.16 — Question 8.** Sketch the following graphs:

a) $y = -(x-4)(x+1)(x+2)^2$

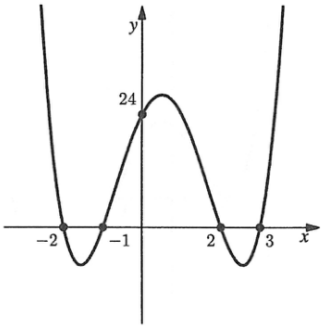
b) $y = (x-1)(3-x)(x+1)^2$

c) $y = -x^2(x-3)^2$

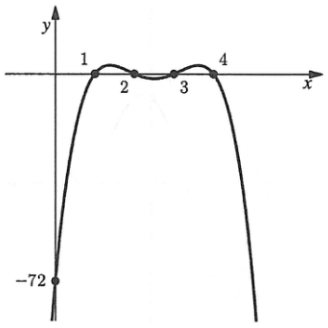
**1.16 — Question 9.** Find the equation of each of the following graphs in the form

$$y = k(x-a)(x-b)(x-c)(x-d)$$

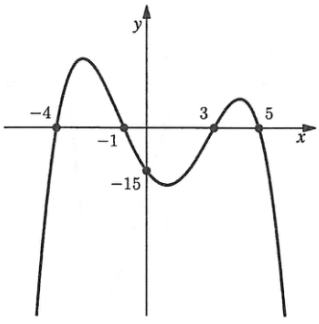
**a)**




**b)**




**c)**



**1.16 — Question 10.** Sketch the following graphs:

a) $y = \frac{1}{2}(x-4)(x+2)(x-2)(x-6)$

b) $y = -\frac{2}{3}(x+3)(x-3)(x-6)^2$

**1.16 — Question 11.** Sketch the following graphs, showing where they intersect the coordinate axes:

a) $y = (2x-1)(x+1)(2x+1)(3x-1)$

b) $y = (x+2)(2x-1)(3x-5)^2$

c) $y = x(3x+4)(5x+2)^2$

d) $y = -x^2(4x-3)(3x-4)$

e) $y = (2x-1)^2(2x+3)^2$

f) $y = -\frac{1}{6}(6x-1)^2(12x+5)^2$

**1.16 — Question 12.** Sketch the following graphs, showing where they intersect the coordinate axes:

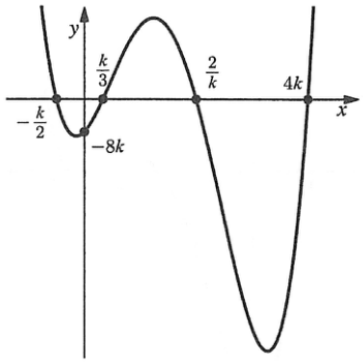
a) $y = (x-2)(x+3)^3$
b) $y = (2x+3)(4x-3)^3$
c) $y = (3-2x)x^3$
d) $y = -\frac{1}{4}(3x+4)(3x-1)^3$

**1.16 — Question 13.** Given that $k > 0$, sketch:

a) $y = x(x-k)(x-2k)(x-3k)$
b) $y = (x+k)(x+3k)(x-2k)(x+2k)$
c) $y = (k-x)(2k+x)(3k-x)^2$
d) $y = (4k-x)(2k+x)^3$

**1.16 — Question 14.** Given that $k$ is a positive constant, find the equation of the following graph in the form

$$y = a(bx + c)(dx + e)(fx + g)(hx + i)$$




**1.16 — Question 15.** Given that $k > 0$, sketch the graphs of

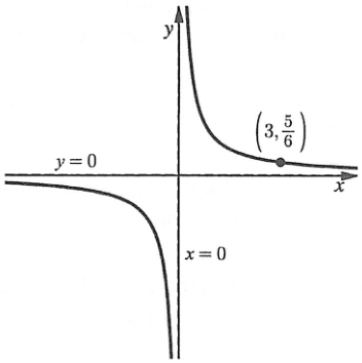
$$y = (x - 3k)(x + 3k)(x - k)^2$$

and

$$y = (3k - x)(3k + x)(k - x)$$

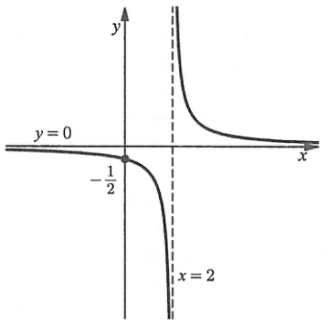
on the same axes, showing clearly where the two graphs cross the coordinate axes and where they intersect.

**1.18 — Question 1.** Find the equation of the following graph in the form $y = \frac{k}{x}$.

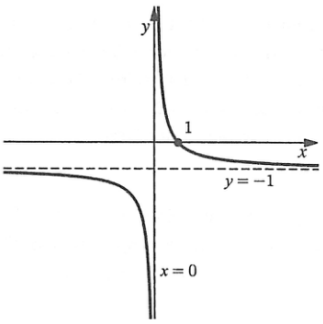



**1.18 — Question 2.** Find the equation of each of the following graphs in the form $y = \frac{1}{x-a} + b$.

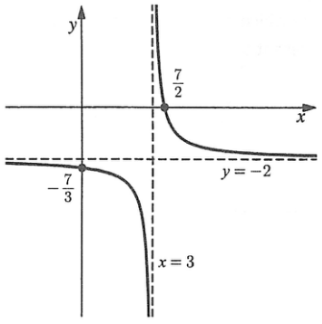
**a)**



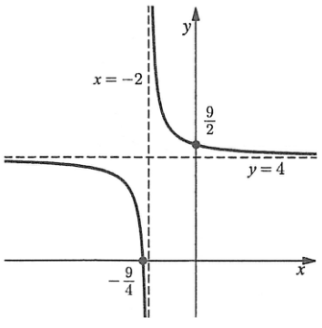

**b)**




**c)**




**d)**



**1.18 — Question 3.** State the equations of the asymptotes of the curve $y = \frac{2}{x-1} + 3$.

**1.18 — Question 4.** Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

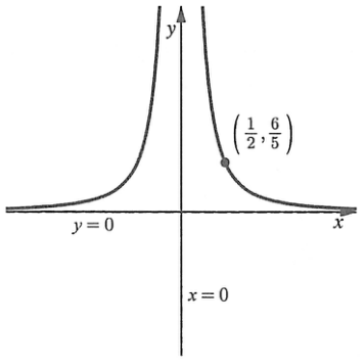
a) $y = \frac{1}{x} - 5$
b) $y = \frac{1}{x+3} - 1$
c) $y = \frac{4}{x-4} + 2$
d) $y = 3 - \frac{1}{x}$
e) $y = 5 - \frac{1}{x+3}$

**1.18 — Question 5.** It is given that $k > 0$. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = \frac{k}{x} - 2k$

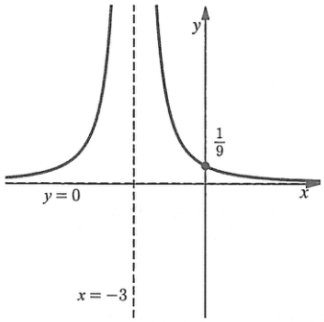
b) $y = k - \frac{1}{x-2k}$

**1.18 — Question 6.** Find the equation of the following graph in the form $y = \frac{k}{x^2}$.

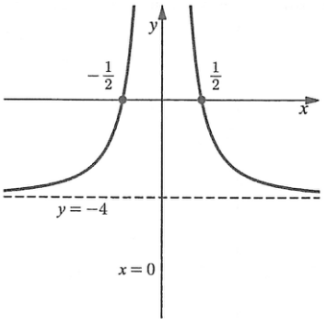


**1.18 — Question 7.** Find the equation of each of the following graphs in the form $y = \frac{1}{(x-a)^2} + b$.

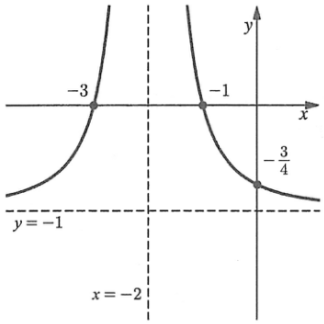
**a)**



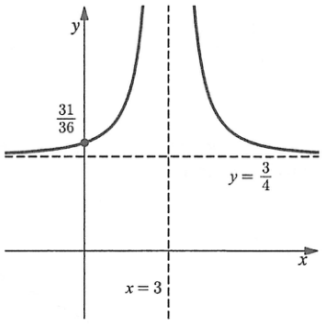

**b)**




**c)**




**d)**




**1.18 — Question 8.** State the equations of the asymptotes of the curve $y = \frac{1}{(x+5)^2} - 2$.

**1.18 — Question 9.** Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = \frac{1}{x^2} + 2$
b) $y = \frac{1}{(x-1)^2} - 4$
c) $y = \frac{4}{(x-4)^2} + 9$
d) $y = 4 - \frac{1}{x^2}$
e) $y = 9 - \frac{1}{(x+2)^2}$

**1.18 — Question 10.** It is given that $k > 0$. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

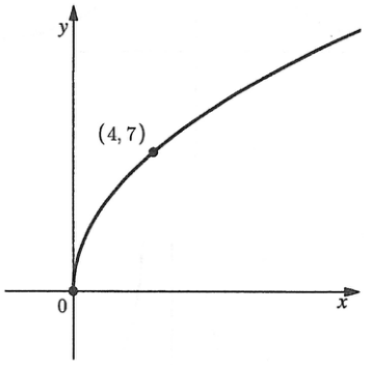
a) $y = \frac{1}{(x-k)^2} + 3k$
b) $y = 4k^2 - \frac{9}{(x+k)^2}$

**1.19 — Question 1.** Sketch the following graphs:

a) $y = \sqrt{x}$

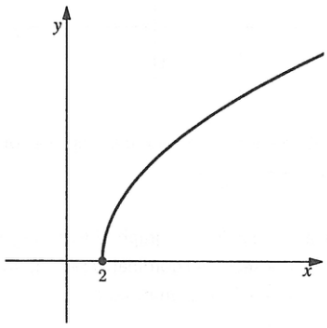
b) $y = -\sqrt{x}$

**1.19 — Question 2.** Find the equation of the following graph in the form $y = k\sqrt{x}$.

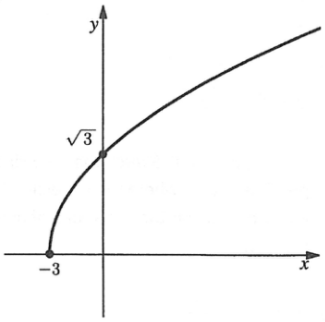


**1.19 — Question 3.** Find the equation of each of the following graphs in the form $y = \sqrt{x - a}$.

**a)**

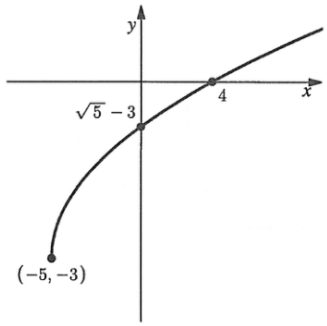



**b)**

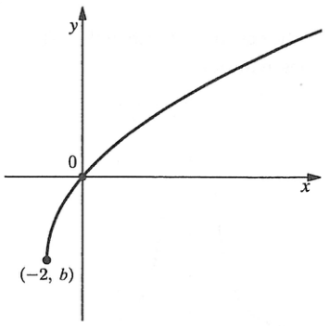



**1.19 — Question 4.** Find the equation of each of the following graphs in the form $y = \sqrt{x - a} + b$.

**a)**




**b)**

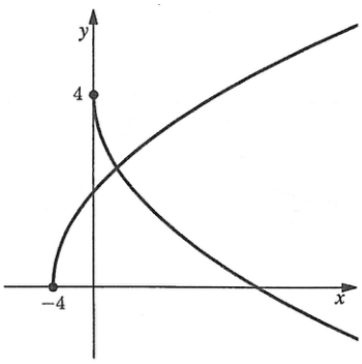


**1.19 — Question 5.** Sketch the following graphs:

a) $y = \sqrt{x-3} - 2$

b) $y = 4 - \sqrt{x+2}$

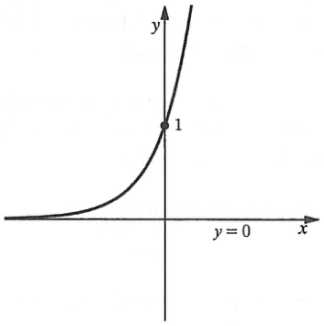
**1.19 — Question 6.** The diagram below shows the graphs $y = 4 - \sqrt{x}$ and $y = \sqrt{x+4}$. Find the exact coordinates of the points of intersection.



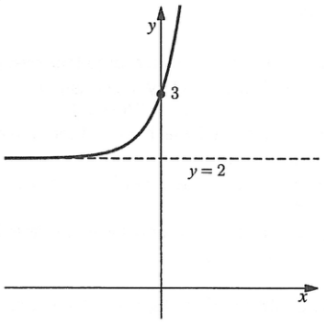
**1.24 — Question 1.** Find the equation of each of the following graphs in the form

$$y = 2^x + b$$

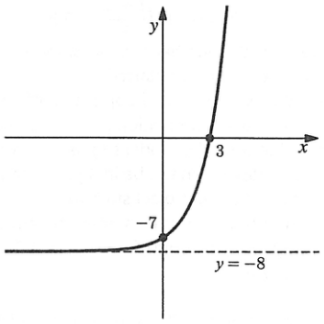
**a)**




**b)**




**c)**




**1.24 — Question 2.** Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

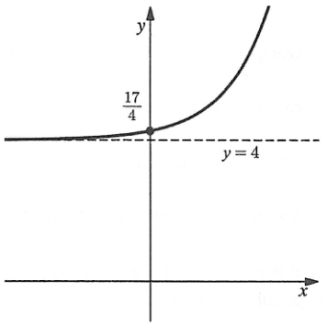
a) $y = 3^x - 9$

b) $y = 4^x + 6$

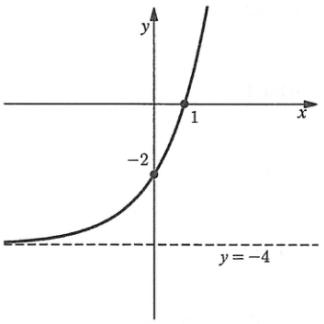
**1.24 — Question 3.** Find the equation of each of the following graphs in the form

$$y = 2^{x-a} + b$$

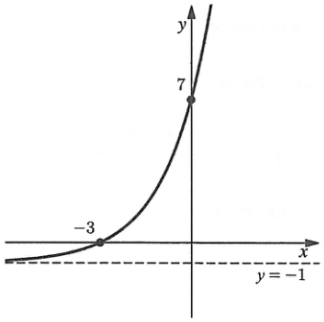
**a)**




**b)**




**c)**




**1.24 — Question 4.** Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

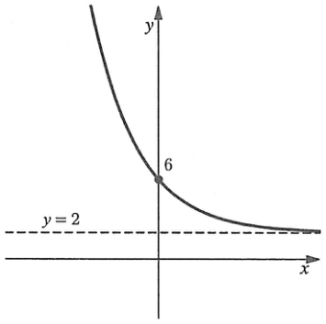
a) $y = 2^{x-4} - 4$

b) $y = 3^{x+2} + 2$

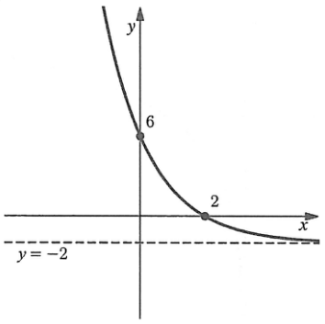
**1.24 — Question 5.** Find the equation of each of the following graphs in the form

$$y = 2^{a-x} + b$$

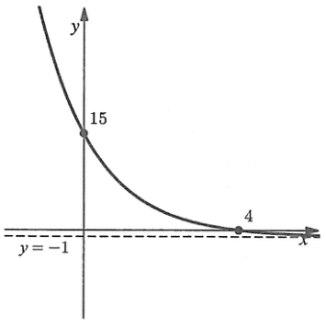
**a)**




**b)**




**c)**




**1.24 — Question 6.** Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

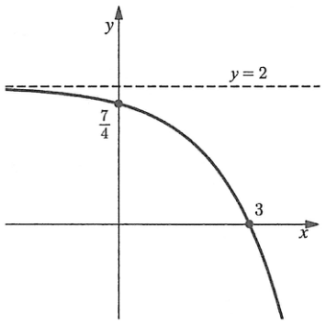
a) $y = 3^{2-x} + 4$

b) $y = 5^{-1-x} - 1$

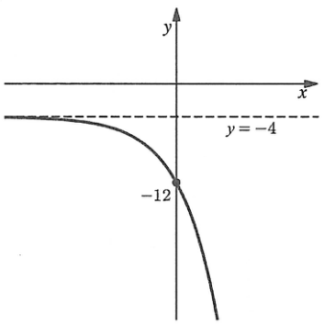
**1.24 — Question 7.** Find the equation of each of the following graphs in the form

$$y = b - 2^{x-a}$$

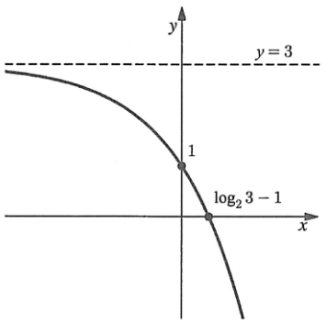
**a)**




**b)**




**c)**




**1.24 — Question 8.** Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

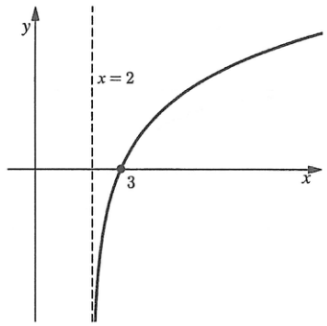
a) $y = 2 - 3^{x-1}$

b) $y = 3 - 2^{1-x}$

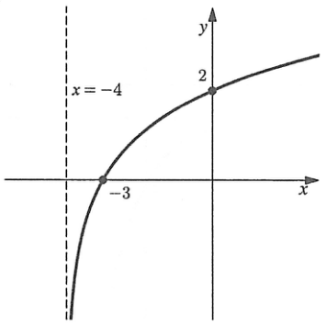
**1.24 — Question 9.** Find the equation of each of the following graphs in the form

$$y = \log_2(x - a)$$

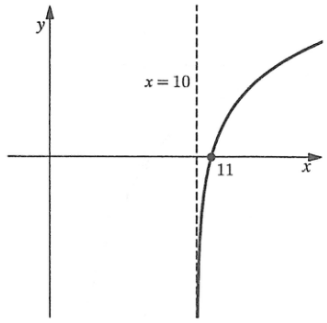
**a)**




**b)**




**c)**




**1.24 — Question 10.** Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

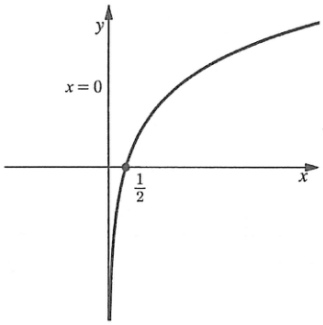
a) $y = \log_3(x - 2)$

b) $y = \log_5(x + 4)$

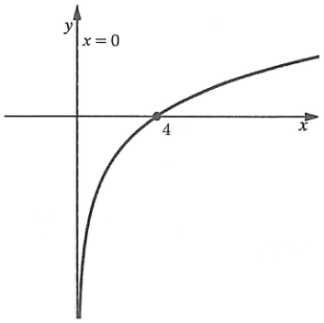
**1.24 — Question 11.** Find the equation of each of the following graphs in the form

$$y = \log_2 x + b$$

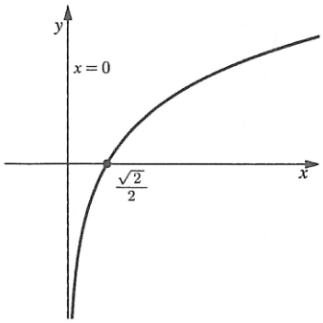
**a)**




**b)**




**c)**




**1.24 — Question 12.** Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

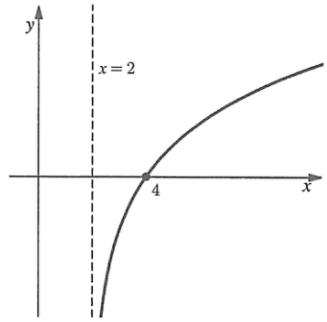
a) $y = \log_5 x - 5$

b) $y = \log_3 x + 4$

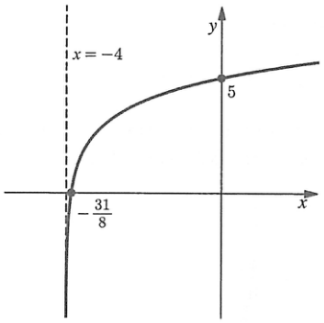
**1.24 — Question 13.** Find the equation of each of the following graphs in the form

$$y = \log_2(x - a) + b$$

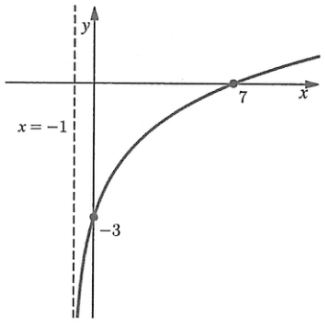
**a)**




**b)**




**c)**



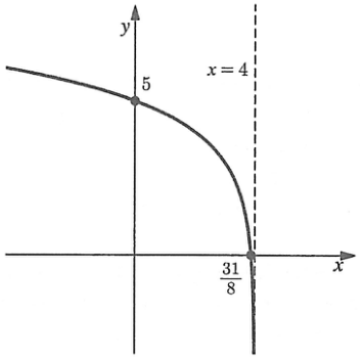

**1.24 — Question 14.** Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = \log_4(x - 2) + 1$

b) $y = \log_3(x + 3) - 3$

**1.24 — Question 15.** Find the equation of the following graph in the form

$$y = \log_2(a - x) + b$$

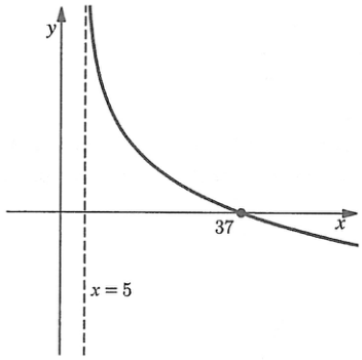



**1.24 — Question 16.** Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = \log_4(5 - x) + 3$
b) $y = \log_5(10 - x) - 2$
c) $y = \log_3(-3 - x) + 1$
d) $y = \log_4(-8 - x) - 2$

**1.24 — Question 17.** Find the equation of the following graph in the form

$$y = b - \log_2(x - a)$$




**1.24 — Question 18.** Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = 3 - \log_3(x - 3)$
b) $y = 2 - \log_4(x + 2)$
c) $y = 1 - \log_5(3 - x)$
d) $y = 2 - \log_3(9 - x)$

**1.24 — Question 19.** Sketch the following pairs of graphs on the same axes, clearly identifying which is which:

a) $y = 2^x$ and $y = 4^x$

b) $y = 3^x$ and $y = 4^{-x}$

c) $y = 2^{-x}$ and $y = 3^{-x}$

**1.24 — Question 20.** Sketch the following pairs of graphs on the same axes, clearly identifying which is which:

a) $y = -5^x$ and $y = -3^x$

b) $y = -2^x$ and $y = 3^x$

c) $y = -4^{-x}$ and $y = -2^{-x}$

**1.24 — Question 21.** Sketch the following pairs of graphs on the same axes, clearly identifying which is which:

a) $y = \log_2 x$ and $y = \log_4 x$

b) $y = \log_5 x$ and $y = \log_3(-x)$

c) $y = \log_6(-x)$ and $y = \log_2(-x)$

**1.24 — Question 22.** Sketch the following pairs of graphs on the same axes, clearly identifying which is which:

a) $y = -\log_4 x$ and $y = -\log_3 x$

b) $y = -\log_8 x$ and $y = \log_4 x$

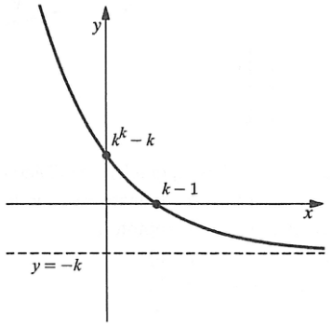
c) $y = -\log_2(-x)$ and $y = -\log_4(-x)$

**1.24 — Question 23.** It is given that $k > 1$. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

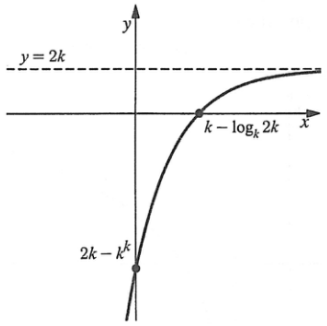
a) $y = k^x$
b) $y = k^x - k$
c) $y = k^{x-k} + k$

**1.24 — Question 24.** It is given that $k > 1$. Write down a suitable equation for each of the following curves.

**a)**




**b)**



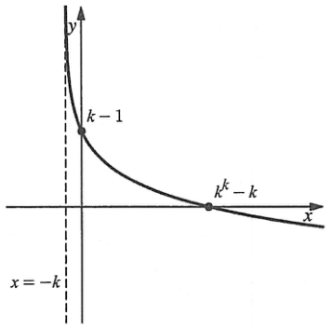
**1.24 — Question 25.** It is given that $k > 1$. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = \log_k x + k$

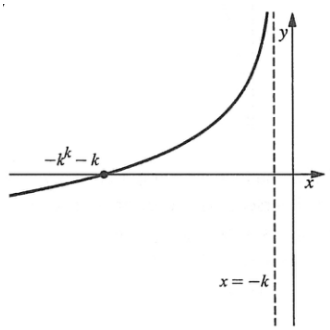
b) $y = \log_k(x + k) - k$

**1.24 — Question 26.** It is given that $k > 1$. Write down a suitable equation for each of the following curves.

**a)**



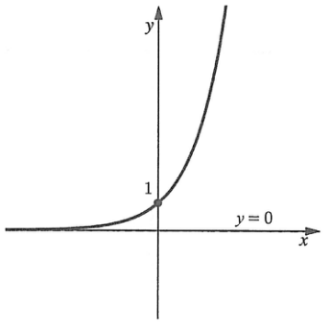

**b)**



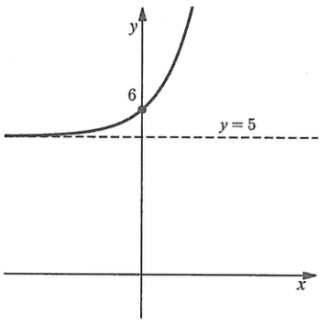
**1.25 — Question 1.** Find the equation of each of the following graphs in the form

$$y = e^x + b$$

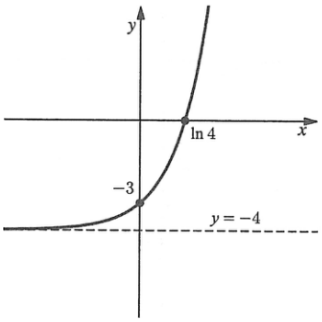
**a)**




**b)**




**c)**




**1.25 — Question 2.** Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

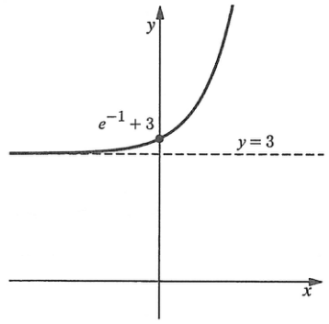
a) $y = e^x - 6$

b) $y = e^x + e$

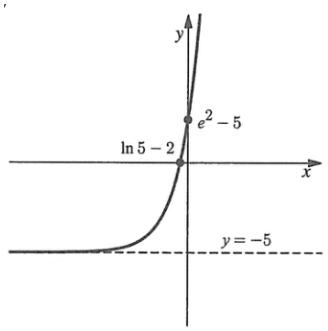
**1.25 — Question 3.** Find the equation of each of the following graphs in the form

$$y = e^{x-a} + b$$

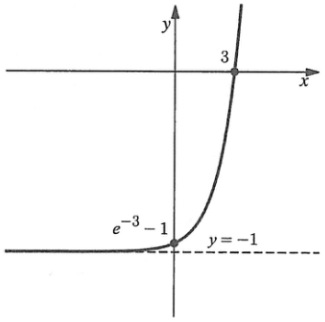
**a)**




**b)**




**c)**




**1.25 — Question 4.** Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

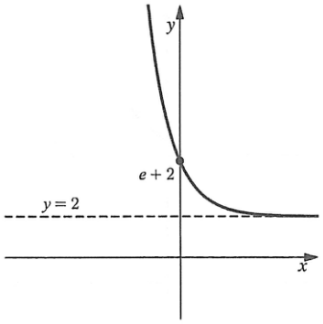
a) $y = e^{x+1} - 10$

b) $y = e^{x-e} - e$

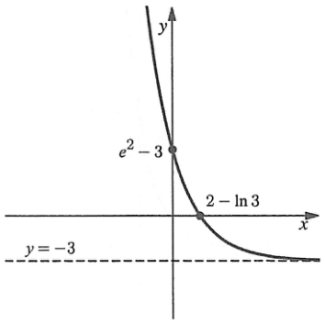
**1.25 — Question 5.** Find the equation of each of the following graphs in the form

$$y = e^{a-x} + b$$

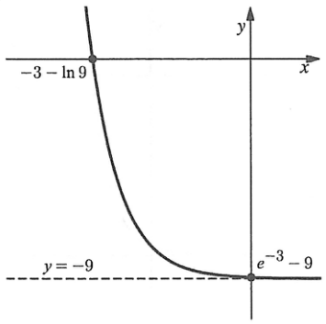
**a)**




**b)**




**c)**




**1.25 — Question 6.** Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

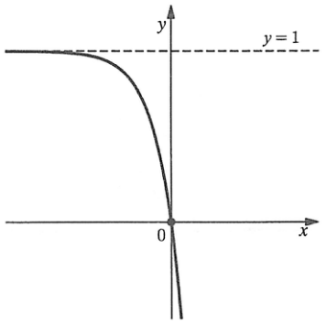
a) $y = e^{5-x} - 20$

b) $y = e^{-1-x} + 20$

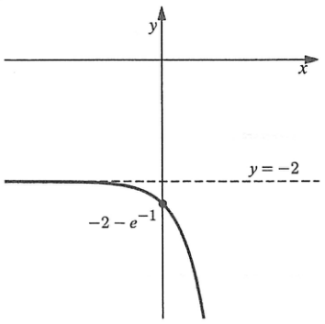
**1.25 — Question 7.** Find the equation of each of the following graphs in the form

$$y = b - e^{x-a}$$

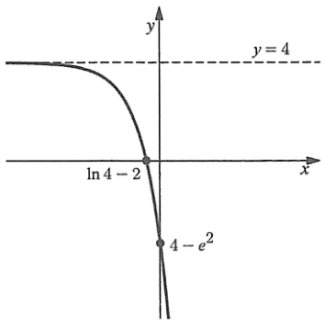
**a)**




**b)**




**c)**




**1.25 — Question 8.** Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

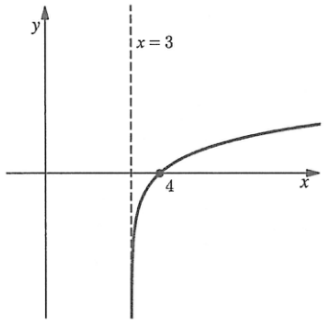
a) $y = 10 - e^{x+4}$

b) $y = 5 - e^{4-x}$

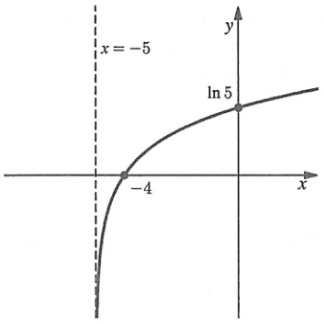
**1.25 — Question 9.** Find the equation of each of the following graphs in the form

$$y = \ln(x - a)$$

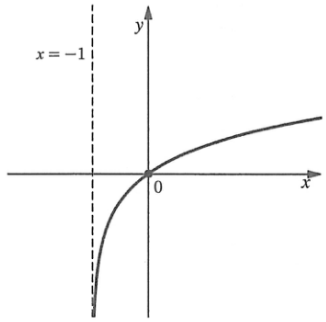
**a)**




**b)**




**c)**




**1.25 — Question 10.** Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

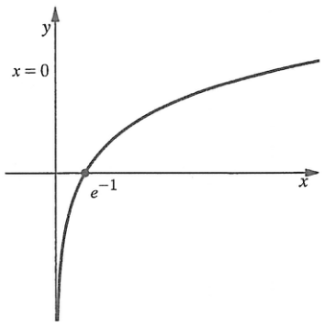
a) $y = \ln(x - 2)$

b) $y = \ln(x + 2)$

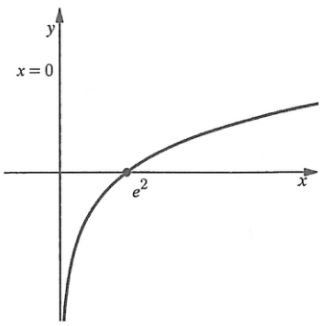
**1.25 — Question 11.** Find the equation of each of the following graphs in the form

$$y = \ln x + b$$

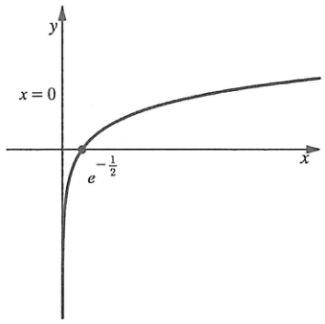
**a)**




**b)**




**c)**




**1.25 — Question 12.** Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

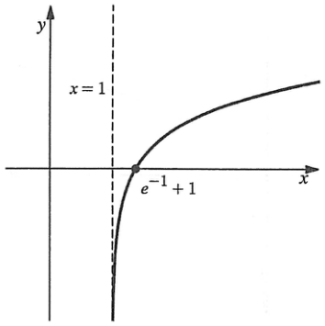
a) $y = \ln x + 3$

b) $y = \ln x - e$

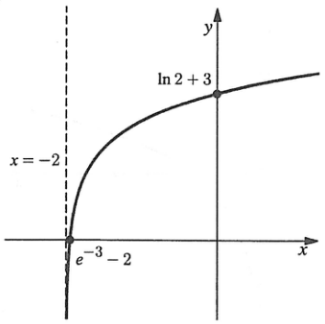
**1.25 — Question 13.** Find the equation of each of the following graphs in the form

$$y = \ln(x - a) + b$$

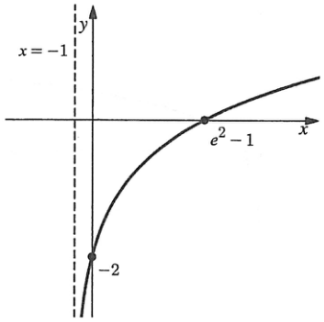
**a)**




**b)**




**c)**



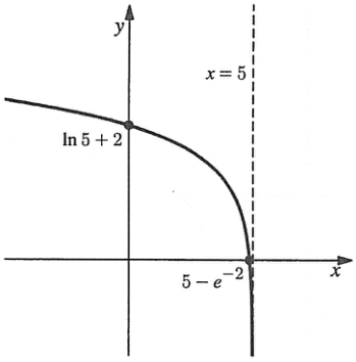

**1.25 — Question 14.** Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = \ln(x - 3) - 3$

b) $y = \ln(x - e) + e$

**1.25 — Question 15.** Find the equation of the following graph in the form

$$y = \ln(a - x) + b$$

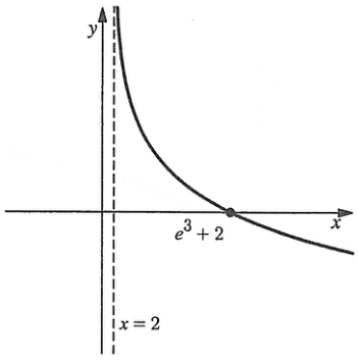



**1.25 — Question 16.** Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = \ln(8 - x) - 1$
b) $y = \ln(1 - x) + 3$
c) $y = \ln(e - x) - e$
d) $y = \ln(-e - x) + 2e$

**1.25 — Question 17.** Find the equation of the following graph in the form

$$y = b - \ln(x - a)$$




**1.25 — Question 18.** Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = 1 - \ln(x - 1)$
b) $y = e - \ln(x + e)$
c) $y = 1 - \ln(2 - x)$
d) $y = \frac{e}{2} - \ln(3e - x)$

**1.25 — Question 19. Prerequisite knowledge for Q19-Q22: Exponentials and Logarithms** Sketch the following pairs of graphs on the same axes, clearly identifying which is which:

a) $y = 2^x$ and $y = e^x$
b) $y = 3^x$ and $y = e^x$
c) $y = e^x$ and $y = e^{-x}$

**1.25 — Question 20.** Sketch the following pairs of graphs on the same axes, clearly identifying which is which:

a) $y = -2^x$ and $y = -e^x$
b) $y = 3^x$ and $y = -e^x$
c) $y = -e^{-x}$ and $y = -e^x$

**1.25 — Question 21.** Sketch the following pairs of graphs on the same axes, clearly identifying which is which:

a) $y = \log_2 x$ and $y = \ln x$

b) $y = \log_3 x$ and $y = \ln x$

c) $y = \log_2(-x)$ and $y = \ln(-x)$

**1.25 — Question 22.** Sketch the following pairs of graphs on the same axes, clearly identifying which is which:

a) $y = -\log_3 x$ and $y = -\ln x$

b) $y = -\log_2 x$ and $y = \ln x$

c) $y = -\log_3(-x)$ and $y = -\ln(-x)$

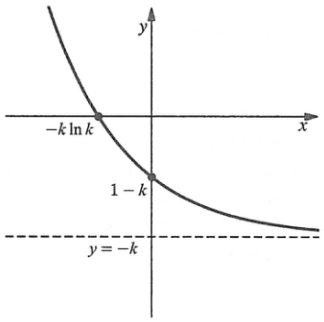
**1.25 — Question 23.** Given that $k > 1$, sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = e^{kx} - 2k$

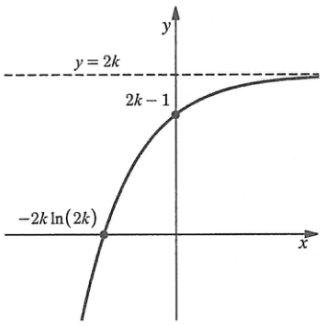
b) $y = k - e^{-kx}$

**1.25 — Question 24.** Given that $k > 1$. Write down a suitable equation for each of the following curves.

**a)**




**b)**



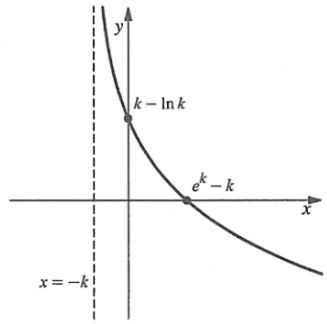
**1.25 — Question 25.** Given that $k > 1$, sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = k - \ln x$

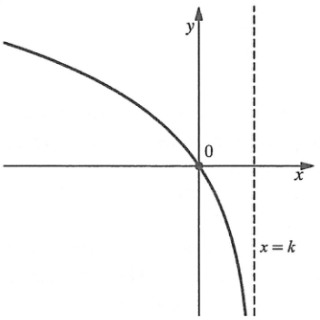
b) $y = k\ln(x + k) - k$

**1.25 — Question 26.** Given that $k > 1$, write down a suitable equation for each of the following curves.

**a)**




**b)**



**1.28 — Question 1. Translations** Describe the transformation that maps $y = \mathrm{f}(x)$ onto each of these:

a) $y = \mathrm{f}(x - 6)$
b) $y = \mathrm{f}(x) + 8$
c) $y = \mathrm{f}(x + 4)$
d) $y = \mathrm{f}(x) - 11$

**1.28 — Question 2.** Describe the transformation that maps $y = x^2$ onto each of these:

a) $y = (x - 9)^2$
b) $y = (x + 5)^2$
c) $y = x^2 + 2$
d) $y = x^2 - 10$
e) $y + 4 = x^2$
f) $y - 8 = x^2$
g) $y = (x + 7)^2 - 3$
h) $y = (x - 1)^2 + 7$

**1.28 — Question 3.** Describe the transformation that maps $y = x^3$ onto each of these:

a) $y = (x + 4)^3$
b) $y = (x - 2)^3$
c) $y = x^3 - 7$
d) $y = x^3 + 15$
e) $y + 9 = x^3$
f) $y = (x - 3)^3 + 8$

**1.28 — Question 4.** Write down the image of $y = 9x + 2$ after a translation by the vector $\begin{pmatrix} -2 \\ 5 \end{pmatrix}$

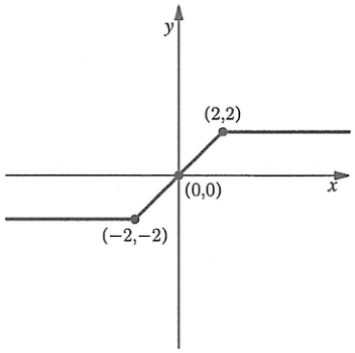
**1.28 — Question 5.** The following functions are to be translated by the vector $\begin{pmatrix} 4 \\ -7 \end{pmatrix}$. Write down the image in each case.

a) $y = 4 - 6x$
b) $y = x^2$
c) $y = x^2 + 3$
d) $y = (x + 2)(x - 8)$
e) $y = (x - 9)(x + 1)$
f) $y = 3^x$
g) $y = e^x - 5$
h) $y = \frac{1}{x}$
i) $y = \frac{1}{x^2}$
j) $y = \sqrt{x}$
k) $y = \cos x$

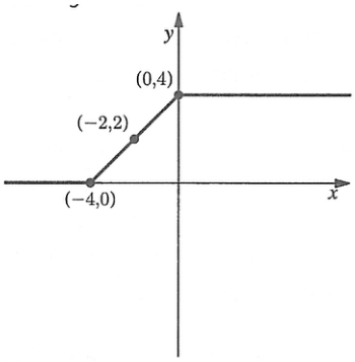
**1.28 — Question 6.** The point $P(7, -5)$ lies on the graph with equation $y = \mathrm{f}(x)$. State the coordinates of the image of P after $y = \mathrm{f}(x)$ is mapped to each of these:

a) $y = \mathrm{f}(x - 8)$
b) $y = \mathrm{f}(x) + 3$
c) $y = \mathrm{f}(x + 12)$
d) $y = \mathrm{f}(x + 1) - 5$

**1.28 — Question 7.** The graph of $y = \mathrm{f}(x)$ is shown in the diagram below.




Write down the equation of the graph shown in the diagram below.




**1.28 — Question 8.** Describe the transformation that maps $y = 3x^4$ onto the image $y = 3(x + 4)^4 - 10$

**1.28 — Question 9.** Describe the transformation that maps $y = x^8 + 7$ onto the image $y = (x - 8)^8$

**1.28 — Question 10.** Describe the transformation that maps $y = \sqrt{x} + 2$ onto the image $y = \sqrt{x + 2}$

**1.28 — Question 11. Prerequisite knowledge for Q11–Q13: Logarithms** Find the translation vector that maps $y = \log_5 x$ onto the image $y = \log_5 \left(\frac{x-4}{25}\right)$.

**1.28 — Question 12.** Find the translation vector that maps $y = \log_3 x$ onto the image $y = \log_3(9(x-2))$.

**1.28 — Question 13.** Fully describe the transformation that maps $y = \ln x$ onto the image $y = \ln\left(\frac{x+2e}{e^5}\right)$.

**1.28 — Question 14.** Find the image of $x^2 + y^2 = 9$ after a translation by the vector $\begin{pmatrix} -7 \\ 5 \end{pmatrix}$.

**1.28 — Question 15.** Find the image of $x^2 + y^2 = 3$ after a translation by the vector $\begin{pmatrix} 2 \\ -8 \end{pmatrix}$.

**1.28 — Question 16.** Find the image of $(x-2)^2 + y^2 = 10$ after a translation by the vector $\begin{pmatrix} -9 \\ -3 \end{pmatrix}$.

**1.28 — Question 17.** Find the image of $(x+10)^2+(y-3)^2=16$ after a translation by the vector $\begin{pmatrix}2\\-1\end{pmatrix}$.

**1.28 — Question 18.** Find the image of $xy+x^2-y=10$ after a translation by the vector $\begin{pmatrix}-3\\4\end{pmatrix}$.

**1.28 — Question 19.** Find the image of $y = x^3 - 2x^2 + 8x + 2$ after a translation by the vector $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$.

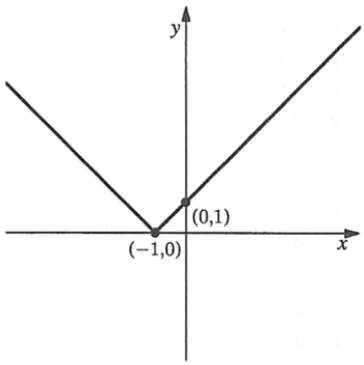
**1.28 — Question 20.** Find the image of $y = 8x^4 - 4x^8$ after a translation by the vector $\begin{pmatrix} 4 \\ -8 \end{pmatrix}$.

**1.28 — Question 21.** Find the image of $y = \frac{3}{x^2}$ after a translation by the vector $\begin{pmatrix} -2 \\ -6 \end{pmatrix}$.

**1.28 — Question 22.** Find the image of $y = -\frac{4}{x-7}$ after a translation by the vector $\begin{pmatrix} 7 \\ 0 \end{pmatrix}$.

**1.28 — Question 23.** Fully describe the transformation that maps $y = 4 \times 4^x$ onto the image $y = 4^{x-5}$.

**1.28 — Question 24.** The diagram below shows the graph of $y = f(x)$. It is given that $f(x)$ is symmetric about $x = -1$.




On separate diagrams, draw a sketch of each of these:

a) $y = f(x + 1)$
b) $y = f(x) - 2$
c) $y = f(x - 4) + 1$
d) $y - 2 = f(x + 2)$

**1.28 — Question 25. Stretches** Describe the transformation that maps $y = f(x)$ onto each of these:

a) $y = f(5x)$
b) $y = \frac{1}{6}f(x)$
c) $y = f(\frac{2}{3}x)$

**1.28 — Question 26.** Describe the transformation that maps $y = x^2 - 3$ onto each of these:

a) $y = 3x^2 - 9$
b) $y = 9x^2 - 3$

**1.28 — Question 27.** Describe the transformation that maps $y = x^3 + 2$ onto each of these:

a) $y = 8x^3 + 2$
b) $y = 8x^3 + 16$
c) $y = \frac{1}{8}x^3 + \frac{1}{4}$
d) $y = \frac{1}{8}x^3 + 2$

**1.28 — Question 28.** Write down the image of $y = 5x + 3$ after:

a) a stretch, parallel to the $x$-axis, scale factor $\frac{1}{4}$
b) a stretch, parallel to the $y$-axis, scale factor $\frac{1}{5}$

**1.28 — Question 29.** The following functions are to be stretched, parallel to the $x$-axis, scale factor 3. Write down the image in each case.

a) $y = 4 - 6x$
b) $y = x^2$
c) $y = x^2 + 3$
d) $y = (x + 2)(x - 8)$
e) $y = (x - 9)(x + 1)$
f) $y = 3^x$
g) $y = e^x - 5$
h) $y = \frac{1}{x}$
i) $y = \frac{1}{x^2}$
j) $y = \sqrt{x}$
k) $y = \cos x$

**1.28 — Question 30.** The following functions are to be stretched, parallel to the $y$-axis, scale factor 3. Write down the image in each case.

a) $y = 4 - 6x$
b) $y = x^2$
c) $y = x^2 + 3$
d) $y = (x + 2)(x - 8)$
e) $y = (x - 9)(x + 1)$
f) $y = 3^x$
g) $y = e^x - 5$
h) $y = \frac{1}{x}$
i) $y = \frac{1}{x^2}$
j) $y = \sqrt{x}$
k) $y = \cos x$

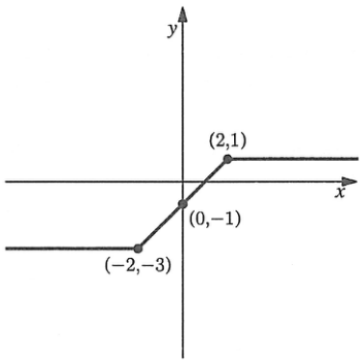
**1.28 — Question 31.** The point $P(-2, 10)$ lies on the graph with equation $y = f(x)$. State the coordinates of the image of $P$ after $y = f(x)$ is mapped to each of these:

a) $y = 6f(x)$
b) $y = f(\frac{1}{2}x)$
c) $y = \frac{2}{5}f(x)$
d) $y = f(6x)$

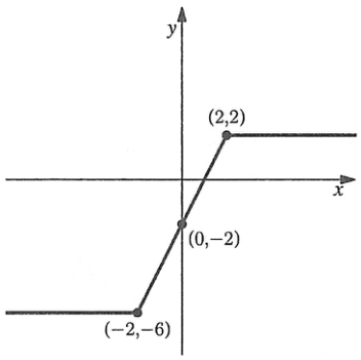
**1.28 — Question 32.** Describe the transformation that maps $y = \sin x$ onto the image $y = \sin 5x$.

**1.28 — Question 33.** Describe the transformation that maps $y = \cos 8x$ onto the image $y = 4\cos 8x$.

**1.28 — Question 34.** The graph of $y = f(x)$ is shown in the diagram below.




Write down the equation of the graph shown in the diagram below.



**1.28 — Question 35.** Describe the transformation that maps $y = (x + 3)^3$ onto the image $y = (\frac{1}{4}x + 3)^3$.

**1.28 — Question 36.** Describe the transformation that maps $y = x^3$ onto the image $y = 27x^3$ as:

a) a stretch parallel to the $y$-axis,

b) a stretch parallel to the $x$-axis.

**1.28 — Question 37.** Describe the transformation that maps $y = \frac{1}{x}$ onto the image $y = \frac{8}{x}$ as:

a) a stretch parallel to the $y$-axis,

b) a stretch parallel to the $x$-axis.

**1.28 — Question 38.** Describe the transformation that maps $y = \frac{1}{x^2}$ onto the image $y = \frac{100}{x^2}$ as:

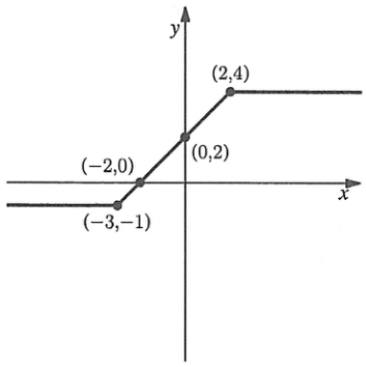
a) a stretch parallel to the $y$-axis,

b) a stretch parallel to the $x$-axis.

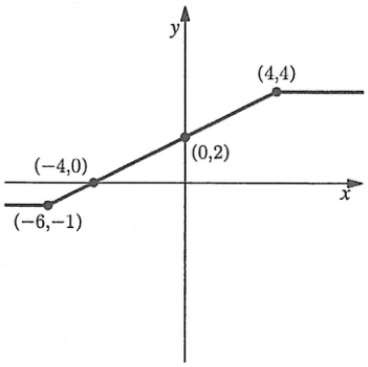
**1.28 — Question 39.** Describe the transformation that maps $x^2 + y^2 = 16$ onto the image $x^2 + 9y^2 = 16$.

**1.28 — Question 40.** Describe the transformation that maps $12x^2 + y^2 = 9$ onto the image $x^2 + y^2 = 9$.

**1.28 — Question 41.** The graph of $y = f(x)$ is shown in the diagram below.




Write down the equation of the graph shown in the diagram below.

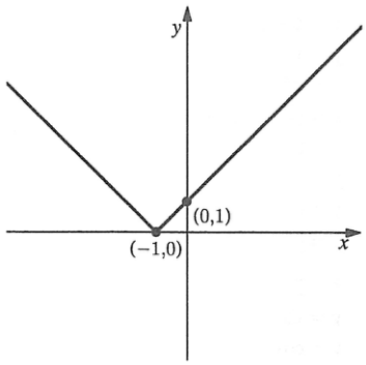



**1.28 — Question 42.** Describe the single transformation that maps $y = 3(x-2)^2 + 5$ onto the image $y = 12(x-1)^2 + 5$.

**1.28 — Question 43.** Describe the single transformation that maps $y = 4(x + 8)^2 - 2$ onto the image $y = 64(x + 2)^2 - 2$.

**1.28 — Question 44.** Describe the single transformation that maps $y = 5(x - \frac{1}{4})^2 + 1$ onto the image $y = \frac{5}{64}(x - 2)^2 + 1$.

**1.28 — Question 45.** The diagram below shows the graph of $y = f(x)$. It is given that $f(x)$ is symmetric about $x = -1$.




On separate diagrams, draw a sketch of each of these:

a) $y = f(2x)$
b) $y = 4f(x)$
c) $3y = f(x)$
d) $y = f(\frac{1}{3}x)$

**1.28 — Question 46. Reflections** Describe the transformation that maps $y = f(x)$ onto each of these:

a) $y = f(-x)$
b) $y = -f(x)$

**1.28 — Question 47.** Describe the transformation that maps $y = (x+3)^2 - 4$ onto each of these:

a) $y = 4 - (x+3)^2$

b) $y = (3-x)^2 - 4$

**1.28 — Question 48. Prerequisite knowledge for Q48:** $e^x$ **&** $\ln x$ Describe the transformation that maps $y = e^x - 2$ onto each of these:

a) $y = e^{-x} - 2$

b) $y = 2 - e^x$

c) $y = \frac{1-2e^x}{e^x}$

**1.28 — Question 49.** Write down the image of $y = 8x - 9$ after:

a) a reflection in the $x$-axis,

b) a reflection in the $y$-axis.

**1.28 — Question 50.** Write down the image of $y = 3 - 6x$ after:

a) a reflection in the $x$-axis,

b) a reflection in the $y$-axis.

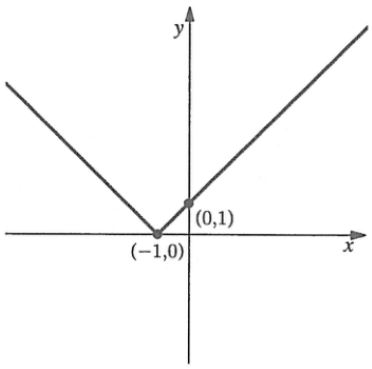
**1.28 — Question 51.** The following functions are to be reflected in the $y$-axis. Identify which (if any) of these functions will remain unchanged:

a) $y = 5 - x$
b) $y = x^2 + 2$
c) $y = x^3$
d) $y = 4^x$
e) $y = \frac{1}{x}$
f) $y = \frac{1}{x^2}$
g) $y = \sin x$
h) $y = \cos x$
i) $y = \frac{e^x + e^{-x}}{2}$

**1.28 — Question 52.** The following functions are to be reflected in the $x$-axis. Identify which (if any) of these functions will remain unchanged:

a) $y = 5 - x$
b) $y = x^2 + 2$
c) $y = x^3$
d) $y = 4^x$
e) $y = \frac{1}{x}$
f) $y = \frac{1}{x^2}$
g) $y = \sin x$
h) $y = \cos x$
i) $y = \frac{e^x + e^{-x}}{2}$

**1.28 — Question 53.** The diagram below shows the graph of $y = f(x)$. It is given that $f(x)$ is symmetric about $x = -1$.




On separate diagrams, draw a sketch of each of these:

a) $y = f(-x)$

b) $y = -f(x)$

**1.28 — Question 54.** The point $P(4, -1)$ lies on the graph with equation $y = f(x)$. State the coordinates of the image of $P$ after $y = f(x)$ is mapped to each of these:

a) $y = -f(x)$

b) $y = f(-x)$

**1.28 — Question 55. Mix** Describe the transformation that would map $y = \sqrt{x}$ onto each of these:

a) $y = \sqrt{x} - 3$
b) $y = \sqrt{2x}$
c) $y = \sqrt{x + 9}$
d) $y = -\sqrt{x}$
e) $y = 8\sqrt{x}$

**1.28 — Question 56.** It is given that $f(x) = 2x^2$ and $g(x) = x - 1$. Show that $y = f(x) + 1$ and $y = g(3x)$ do not intersect.

**1.28 — Question 57.** It is given that $f(x) = (x-1)^2 - 1$ and $g(x) = 3x$. Show that $y = 2f(x)$ and $y = g(x-2)$ intersect at two distinct points.

**1.28 — Question 58.** Describe a single transformation that will map $y = \frac{16}{x+1}$ onto the image $y = \frac{9}{x+1}$.

**1.28 — Question 59.** It is given that $f(x) = \frac{1}{x^2}$. State the asymptotes of the following graphs:

a) $y = f(x + 2)$
b) $y = -f(x)$
c) $y = f(x - 5) + 4$
d) $y = f(\frac{1}{3}x)$

**1.28 — Question 60.** It is given that $y = f(x)$ is a polynomial of degree 3. It is also given that $f(-2) = 0$, $f(5) = 0$ and $f(7) = 0$. $y = f(x)$ is mapped to the image $y = g(x)$. It is now given that $g(a) = 0$, $g(b) = 0$ and $g(c) = 0$. State the values of $a$, $b$ and $c$ when:

a) $g(x) = f(x - 2)$
b) $g(x) = f(x + 10)$
c) $g(x) = 4f(x)$
d) $g(x) = f(2x)$
e) $g(x) = f(\frac{1}{3}x)$
f) $g(x) = f(-x)$

**1.28 — Question 61.** It is given that $f(x) = (x-2)(x+4)(x+5)$.

a) Solve $f(x) = 0$
b) Solve $f(x-2) = 0$
c) Solve $f(2x) = 0$
d) Solve $f(\frac{1}{3}x) = 0$
e) Solve $f(-x) = 0$
f) Solve $f(x+15) = 0$

**1.28 — Question 62.** It is given that $f(x) = \frac{8}{x-6}$. State the equation of the image of $f(x)$ after each of these transformations:

a) A translation by the vector $\begin{pmatrix} -3 \\ 0 \end{pmatrix}$
b) A stretch, parallel to the $y$-axis, scale factor $\frac{1}{2}$
c) A reflection in the $y$-axis
d) A stretch, parallel to the $x$-axis, scale factor $\frac{1}{4}$

**1.28 — Question 63.** It is given that $y = f(x)$ is a polynomial of degree 3. It is also given that $f(-3) = 0$, $f(6) = 0$ and $f(8) = 0$. $y = f(x)$ is mapped to the image $y = g(x)$, where $g(x) = f(x - k)$ and $k$ is a constant. It is also given that $y = g(x)$ passes through $(1, 0)$. State the possible values of $k$.

**1.28 — Question 64.** It is given that $y = f(x)$ is a polynomial of degree 3. It is also given that $f(-8) = 0$, $f(-2) = 0$ and $f(3) = 0$. $y = f(x)$ is mapped to the image $y = g(x)$, where $g(x) = f(kx)$, where $k$ is a constant. It is also given that $y = g(x)$ passes through $(-1, 0)$. State the possible values of $k$.

**1.28 — Question 65.** It is given that $f(x) = 4^x$. The graph of $y = f(x)$ is mapped onto the image $y = g(x)$, where $g(x) = 4^{x-3}$.

a) Fully describe the transformation using a translation.

b) Fully describe the transformation using a stretch.

**1.28 — Question 66.** Fully describe the single transformation that maps $y = x^2$ onto the curve $y = x^2 + 8x - 3$.

**1.28 — Question 67.** Fully describe the single transformation that maps $y = 2x^2$ onto the curve $y = 2x^2 - 8x - 5$.

**1.28 — Question 68.** It is given that $f(x) = 3x^2 - 2x + 1$. Find the new equation of the curve in the form $y = ax^2 + bx + c$ after each of these transformations:

a) A reflection in the $y$-axis

b) A translation by the vector $\begin{pmatrix} 3 \\ -6 \end{pmatrix}$

c) A stretch, parallel to the $x$-axis, scale factor $\frac{1}{3}$

**1.28 — Question 69.** It is given that $f(x) = 2x^4 - 7x^3 - 4x^2$.

a) Solve $f(x) = 0$

It is given that $g(x) = 2(x+3)^4 - 7(x+3)^2 - 4(x+3)^2$.

b) Use your answer to a) to solve $g(x) = 0$

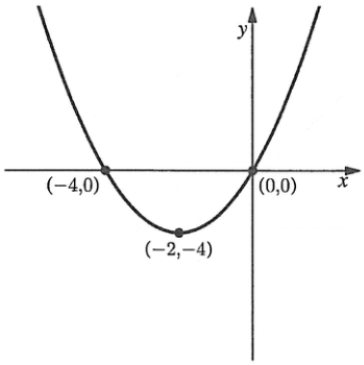
It is given that $h(x) = 162x^4 - 189x^3 - 36x^2$.

c) Use your answer to a) to solve $h(x) = 0$

It is given that $m(x) = 2x^4 + 7x^3 - 4x^2$.

d) Use your answer to a) to solve $m(x) = 0$

**1.28 — Question 70.** The diagram below shows the graph of $y = f(x)$. It is given that $f(x)$ is symmetric about $x = -2$.




On separate diagrams, draw a sketch of each of these:

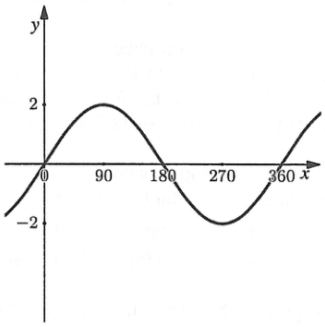
a) $y = -f(x)$

b) $y = f(x-3)$

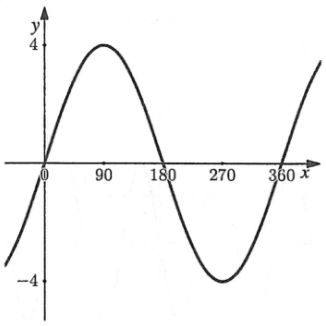
c) $y = \frac{5}{2}f(x)$

**1.41 — Question 1.** Find the equation of each of the following graphs in the form $y = a\sin x$.

**a)**

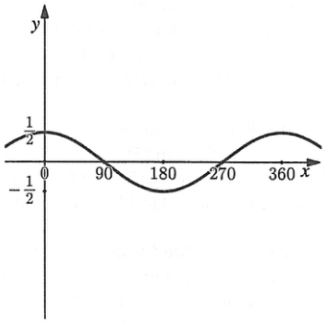



**b)**

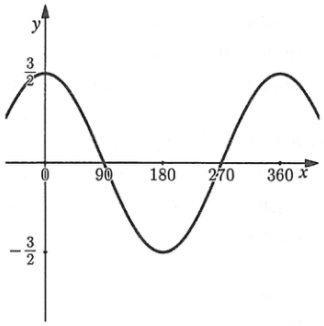



**1.41 — Question 2.** Find the equation of each of the following graphs in the form $y = a\cos x$.

**a)**




**b)**



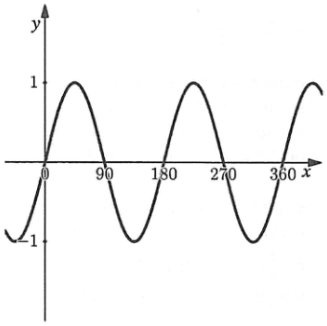
**1.41 — Question 3.** Sketch the following graphs for $0° \leq x \leq 360°$:

a) $y = \frac{6}{5}\sin x$

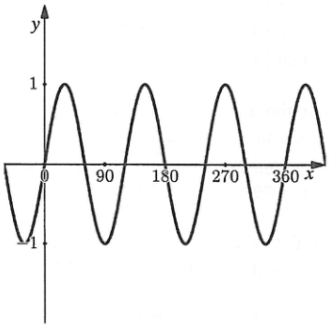
b) $y = \frac{2}{3}\cos x$

**1.41 — Question 4.** Find the equation of each of the following graphs in the form $y = \sin ax$.

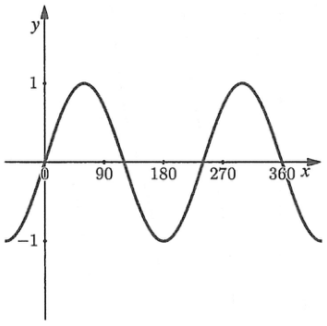
**a)**



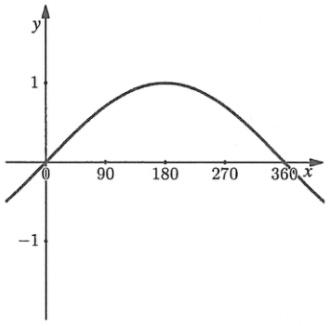

**b)**




**c)**



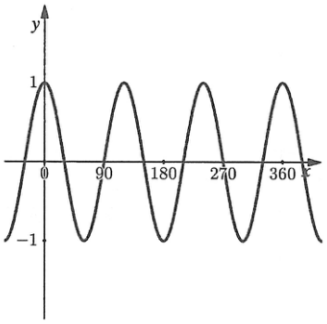

**d)**



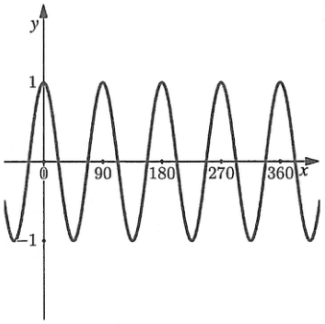
**1.41 — Question 5.** Sketch the graph of $y = -2\sin x$ for $0° \leq x \leq 360°$.

**1.41 — Question 6.** Find the equation of each of the following graphs in the form $y = \cos ax$.

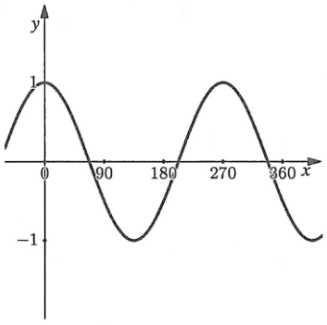
**a)**



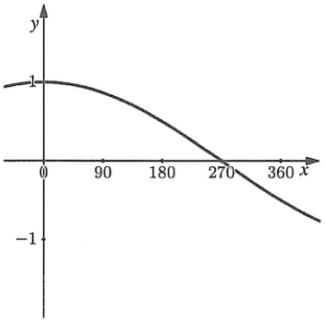

**b)**




**c)**




**d)**

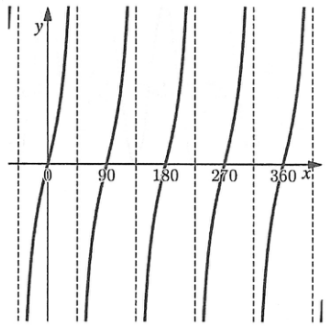


**1.41 — Question 7.** Sketch the graph of $y = -3\cos x$ for $0° \leq x \leq 360°$.

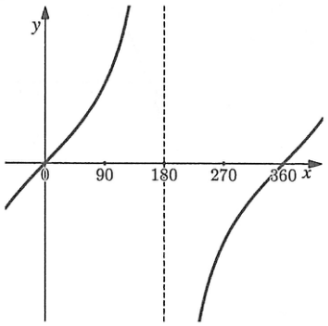
**1.41 — Question 8.** Sketch the graph of $y = \sin(-2x)$ for $0° \leq x \leq 360°$.

**1.41 — Question 9.** Find the equation of each of the following graphs in the form $y = \tan ax$.

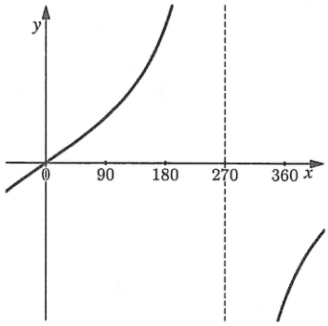
**a)**



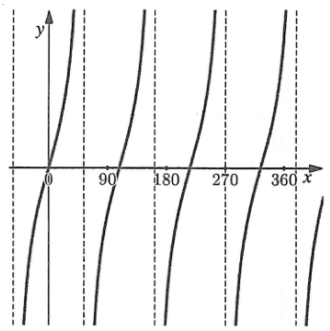

**b)**




**c)**




**d)**



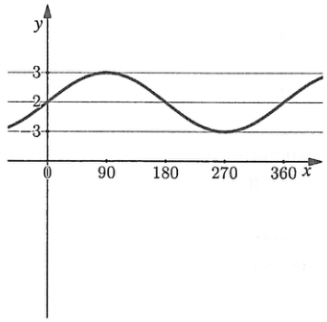

**1.41 — Question 10.** Sketch the following graphs for $0° \leq x \leq 360°$:

a) $y = \tan \frac{3}{2}x$

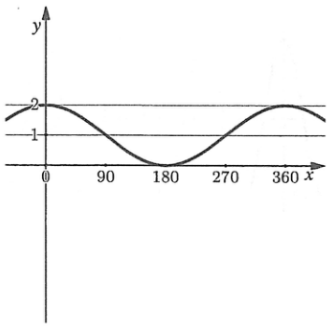
b) $y = \tan \frac{2}{3}x$

**1.41 — Question 11.** Find the equation of each of the following graphs in either the form $y = a + \sin x$ or the form $y = a + \cos x$.

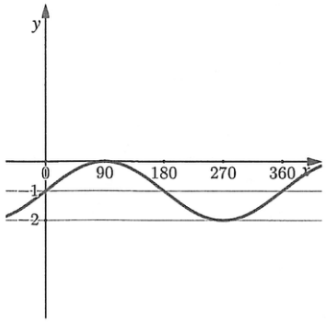
**a)**



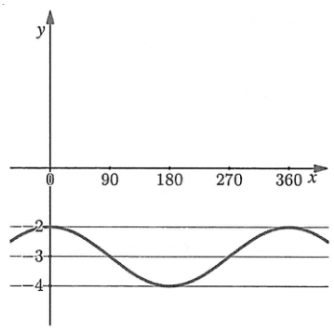

**b)**




**c)**

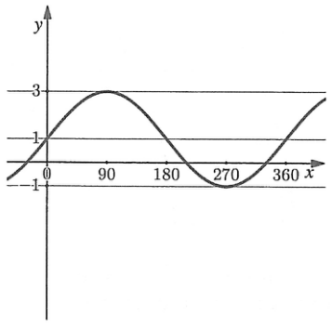



**d)**

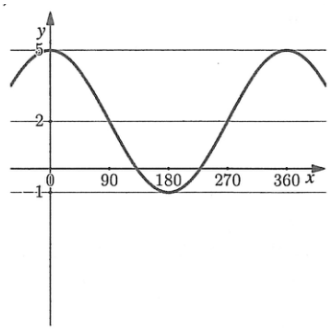



**1.41 — Question 12. Prerequisite knowledge for Q12–Q13: Combinations of Graph Transformations**
Find the equation of each of the following graphs in the form $y = a + b\sin x$ or the form $y = a + b\cos x$.

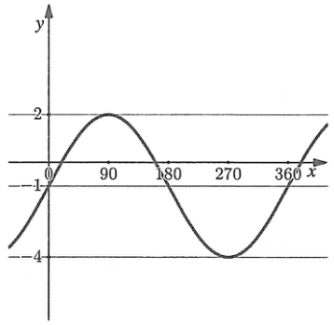
**a)**



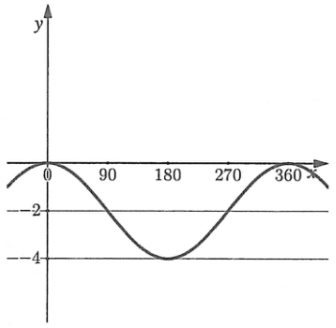

**b)**




**c)**




**d)**



**1.41 — Question 13.** Sketch the following graphs for $0° \leq x \leq 360°$:

a) $y = 1 + 3\sin x$

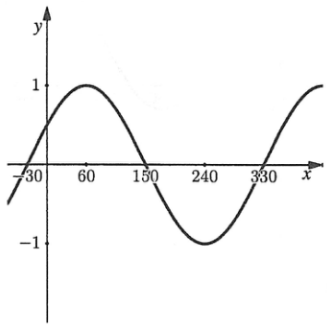
b) $y = -1 + 4\cos x$

c) $y = 2 - 2\sin x$

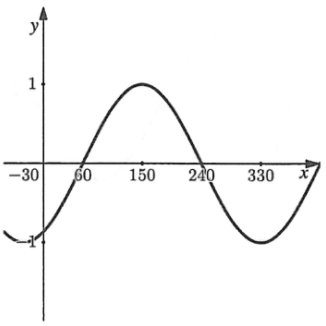
d) $y = -3 - 2\cos x$

**1.41 — Question 14.** Find the equation of each of the following graphs in the form $y = \sin(x + a)$.

**a)**

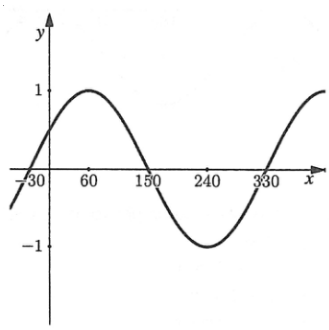



**b)**

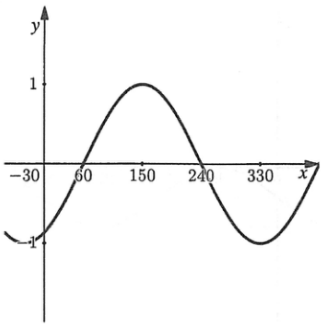


**1.41 — Question 15.** Find the equation of each of the following graphs in the form $y = \cos(x + a)$.

**a)**

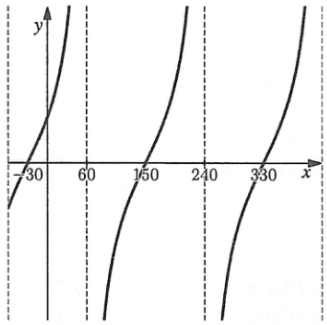



**b)**

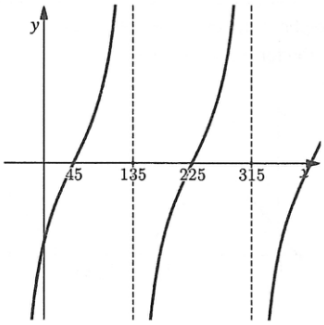



**1.41 — Question 16.** Find the equation of each of the following graphs in the form $y = \tan(x + a)$.

**a)**




**b)**



**1.41 — Question 17.** It is given that $k$ is a constant where $k > 1$. Sketch the following graphs for $0° \leq x \leq 360°$:

a) $y = k + \sin x$

b) $y = -k + \cos x$

**1.41 — Question 18.** Sketch the following graphs for $0° \leq x \leq 360°$:

a) $y = \sin(x - 90°)$

b) $y = \cos(x + 60°)$

c) $y = \tan(x + 90°)$

d) $y = \sin(x + 120°)$

e) $y = \cos(x - 40°)$

f) $y = \tan(x - 50°)$

**1.41 — Question 19.** It is given that $k$ is a constant where $0° < k < 90°$. Sketch the following graphs for $0° \leq x \leq 360°$:

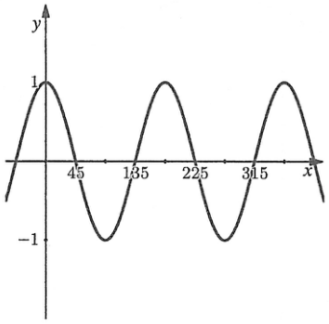
a) $y = \sin(x + k)$

b) $y = \cos(x - k)$

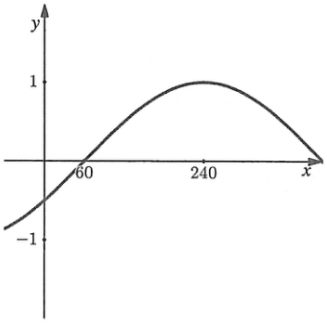
**1.41 — Question 20. Prerequisite knowledge for Q20–Q25: Combinations of Graph Transformations**

Find the equation of each of the following graphs in the form $y = \sin(ax + b)$.

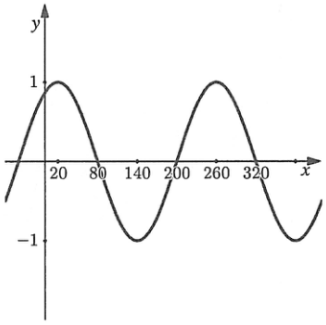
**a)**




**b)**

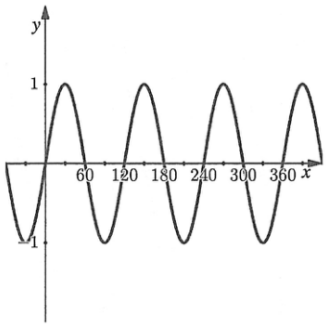



**c)**

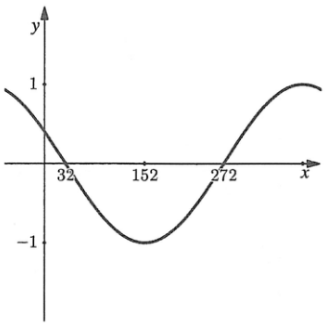


**1.41 — Question 21.** Find the equation of each of the following graphs in the form $y = \cos(ax + b)$.

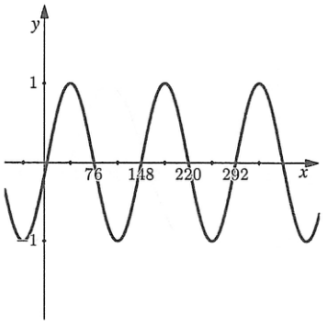
**a)**




**b)**

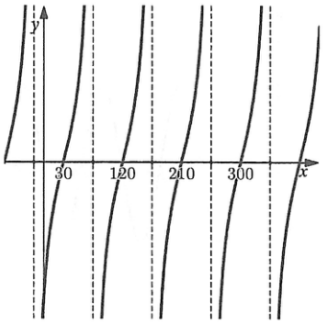



**c)**

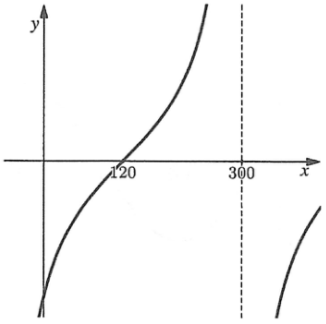



**1.41 — Question 22.** Find the equation of each of the following graphs in the form $y = \tan(ax + b)$.

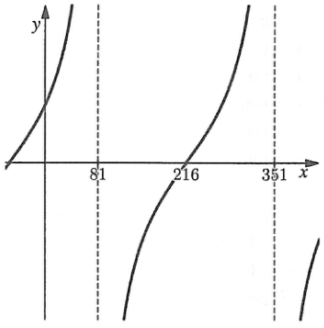
**a)**




**b)**

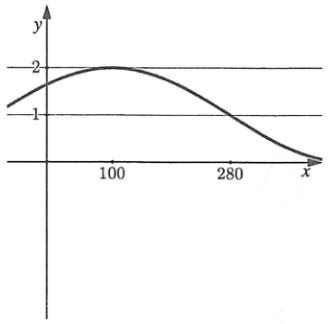



**c)**

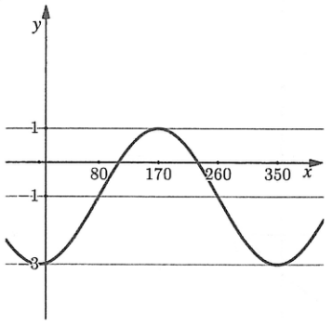


**1.41 — Question 23.** Find the equation of each of the following graphs in the form $y = a + b\sin(cx + d)$ where $-180° < d < 180°$.

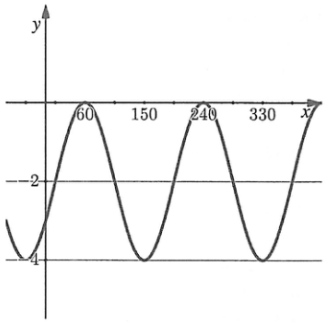
**a)**



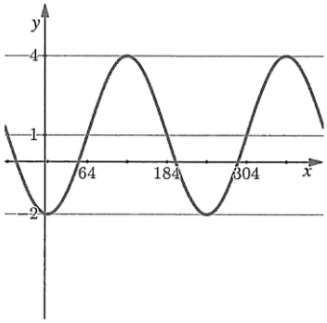

**b)**




**c)**

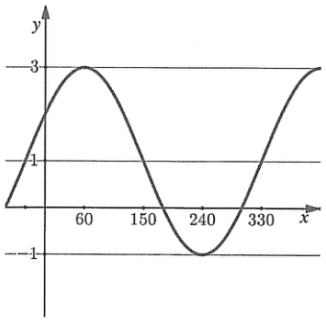



**d)**

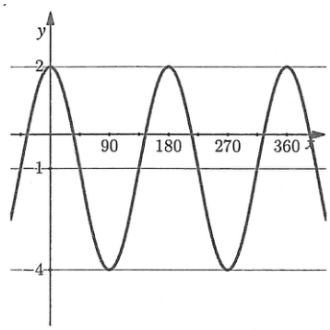



**1.41 — Question 24.** Find the equation of each of the following graphs in the form $y = a + b\cos(cx + d)$ where $-180° < d < 180°$.

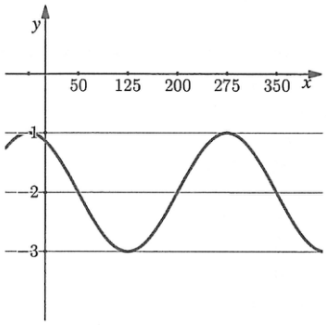
**a)**



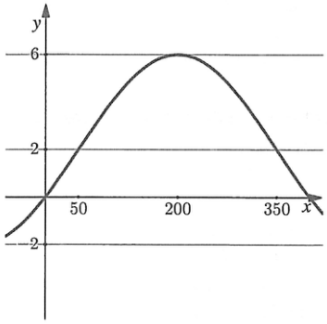

**b)**




**c)**




**d)**



**1.41 — Question 25.** It is given that $a$ and $b$ are positive constants. Sketch the following graphs for $0° \leq x \leq 360°$:

a) $y = a + b\sin x$ where $a > b$

b) $y = a + b\sin x$ where $a = b$

c) $y = a + b\sin x$ where $a < b$